

Computer Networks

21CSC302J

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Units to be Covered

Unit-1 - Introduction to Networks

Unit-2 – Addressing

Unit-3 – Routing

Unit-4 – Medium Access Control

Unit-5 – Transport and Application Layer Protocols

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Unit – I – Introduction to Networks

Topics to be Covered

1.1 Network Types: LAN, MAN, WAN, PAN

1.2 Network Topology : BUS, STAR, RING, MESH, HYBRID

1.3 Switching : Circuit Switching, Packet Switching

1.4 OSI Layered Architecture

1.5 TCP/IP Model

1.6 Physical Layer Overview - Latency, Bandwidth, Delay

1.7 Guided Media : Twisted pair, Coaxial cable, Fiber optic cable

1.8 Unguided Media : Radio waves, Microwaves, Infrared.

Networks

❖ A **network** is a group of **nodes (devices)** that are **connected** to share or exchange information.

❖ A node is a device that is capable of sending and receiving data.

Examples: Computer, Laptop, Router, Switch, Smartphone, etc.

❖ Data means information.

❖ It can be in any form (text, audio, video, image) that is understood by sender and receiver.

❖ Data Communication is the exchange of data between two devices using a transmission medium.

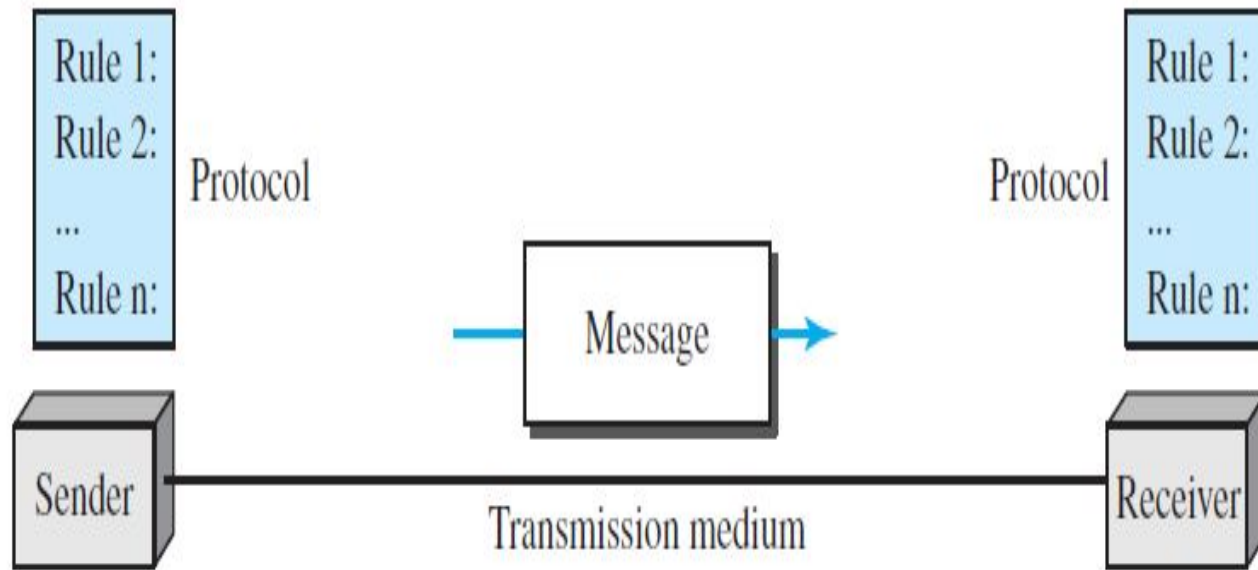
Example: Wire cable, Optical fiber, Wireless signal (Wi-Fi)

❖ To make communication possible, some key components need to work together.

❖ These components ensure that the message is sent, transmitted, and received accurately between the sender and the receiver.

Networks

- ❖ The following diagram image illustrates the basic components of a data communication system.



- ❑ **Sender:** The device or system that initiates and sends the message (e.g., a computer).
- ❑ **Receiver:** The device or system that receives the message (e.g., another computer or a server).
- ❑ **Message:** The actual data or information being transmitted from the sender to the receiver.
- ❑ **Transmission Medium:** The physical or logical path through which the message travels (e.g., cables, optical fiber, or wireless signals).
- ❑ **Protocol:** A set of rules and conventions (Rule 1, Rule 2, ..., Rule n) followed by both the sender and receiver to ensure proper and successful communication.

- ❖ Communication between two devices can be simplex, half-duplex, or full-duplex

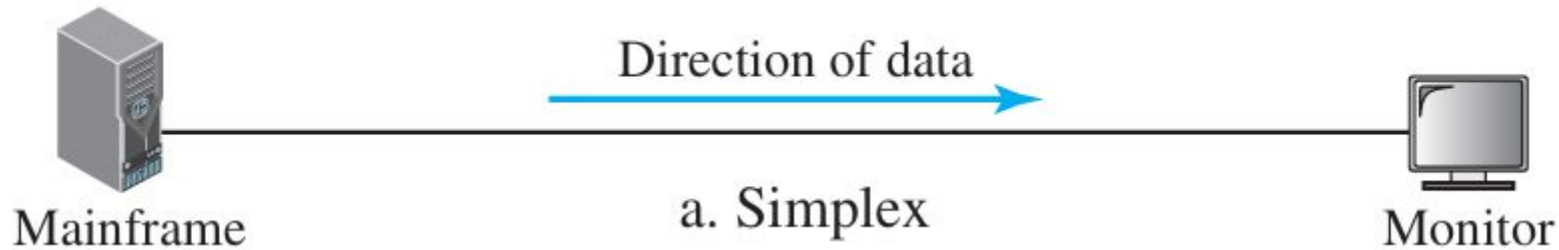
Networks

1. **Simplex**

- ❖ The communication is unidirectional, as on a one-way street.
- ❖ Only one of the two devices on a link can transmit; the other can only receive.

Examples: Keyboards and traditional monitors.

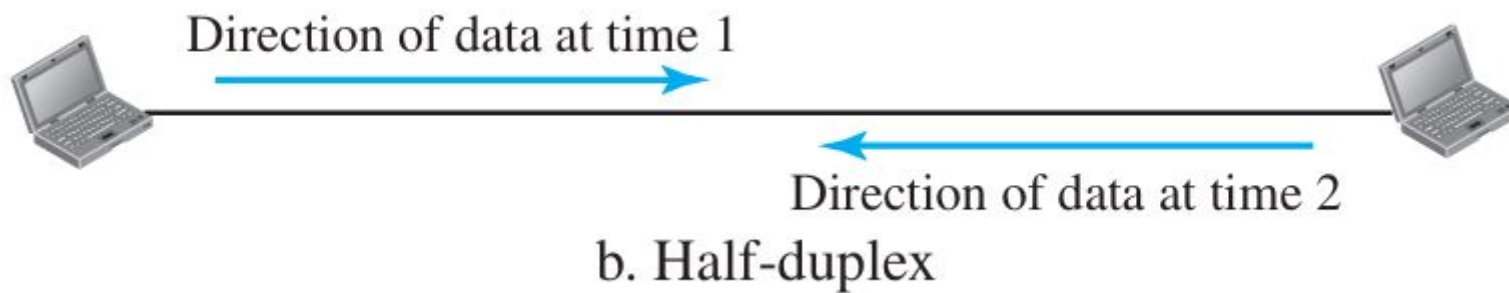
- ❖ The key board can only introduce input; the monitor can only accept output.
- ❖ The simplex mode can use the entire capacity of the channel to send data in one direction.



Networks

2. Half-Duplex

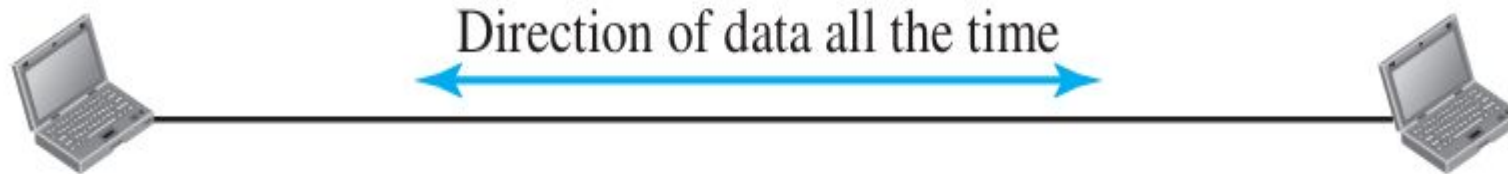
- ❖ In half-duplex mode, each station can both transmit and receive, but not at the same time.
- ❖ When one device is sending, the other can only receive, and vice versa.
- ❖ Walkie-talkies and CB (citizens band) radios are both half-duplex systems.



Networks

3. Full-Duplex

- ❖ In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously.
- ❖ In full-duplex mode, data can be sent and received at the same time.
- ❖ This is done either using two separate paths or by splitting the channel's capacity for both directions.



c. Full-duplex

1.1 Network Types

- ❖ A **computer network** is a group of connected devices that can communicate and share data, resources, and information. Networks help users share files, printers, and access the internet.
- ❖ Based on the size of the area they cover, networks are categorized into different types.
 - **Personal Area Network(PAN)**
 - **Local Area Network(LAN)**
 - **Metropolitan Area Network(MAN)**
 - **Wide Area Network(WAN)**
- ❖ Each type of network serves a specific purpose and is suitable for different environments — from personal use at home to connecting computers across the globe.

1.1 Network Types

1. Personal Area Network (PAN)

- ❖ A Personal Area Network (PAN) is a small network used to connect devices within a short range—typically around 10 meters—around a single person.
- ❖ It enables communication and data sharing between personal devices.
- ❖ It Allows data, files, internet, or device sharing between personal gadgets (e.g., sharing a mobile hotspot with a laptop).
- ❖ Examples: Bluetooth, Infrared.
- ❖ **Speed:** Faster than traditional wired serial connections but slower than Wi-Fi or LAN.
- ❖ **Security:** It is secure enough for personal use, but vulnerable if not configured properly
- ❖ **Advantages:**
 - Easy to set up
 - Good for personal use
- ❖ **Disadvantages:**
 - Limited range
 - Not suitable for large-scale communication



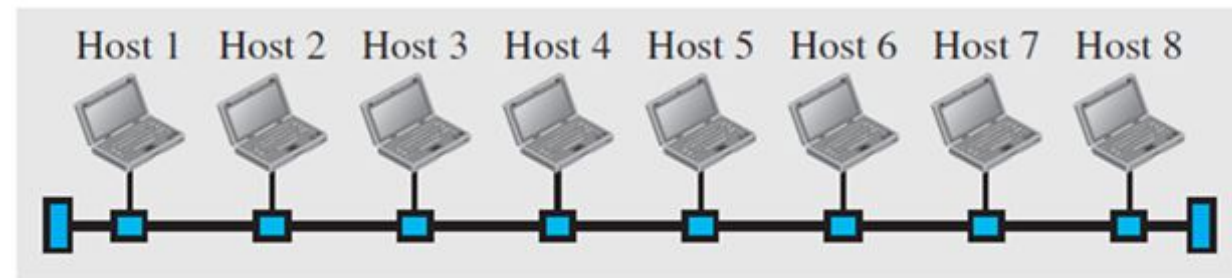
1.1 Network Types

2. Local Area Network (LAN)

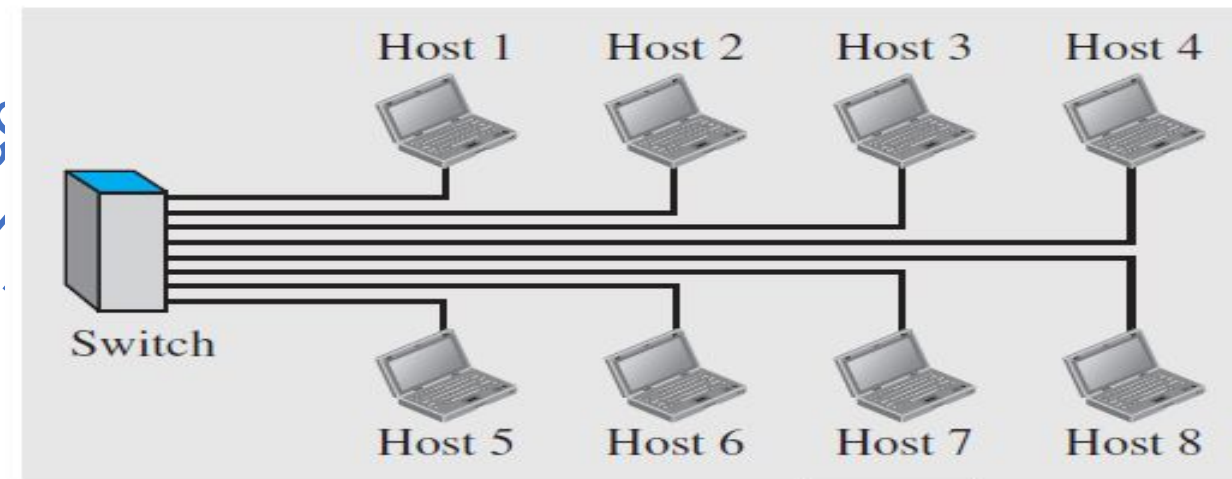
- ❖ Connecting computers within a small area like an office, home, or school — typically up to 1 km in range.
- ❖ **Speed:** 10 Mbps to 10 Gbps
- ❖ **Connection:** Wired (Twisted pair, coaxial, fiber) or Wireless
- ❖ **Protocol:** Uses rules for communication called protocols
- ❖ **Topologies:** Bus, Ring, Star

Types of LAN:

- Cable-based LAN
- PBX (Private Branch Exchange)
- Hierarchical LAN



a. LAN with a common cable



b. LAN with a switch

Advantages:

- ✓ High speed and low cost
- ✓ Easy file and hardware sharing
- ✓ Secure and efficient

Disadvantages:

- ✓ Setup cost may be high
- ✓ Needs maintenance
- ✓ Server or cable failure can stop network

1.1 Network Types

3. Metropolitan Area Network (MAN)

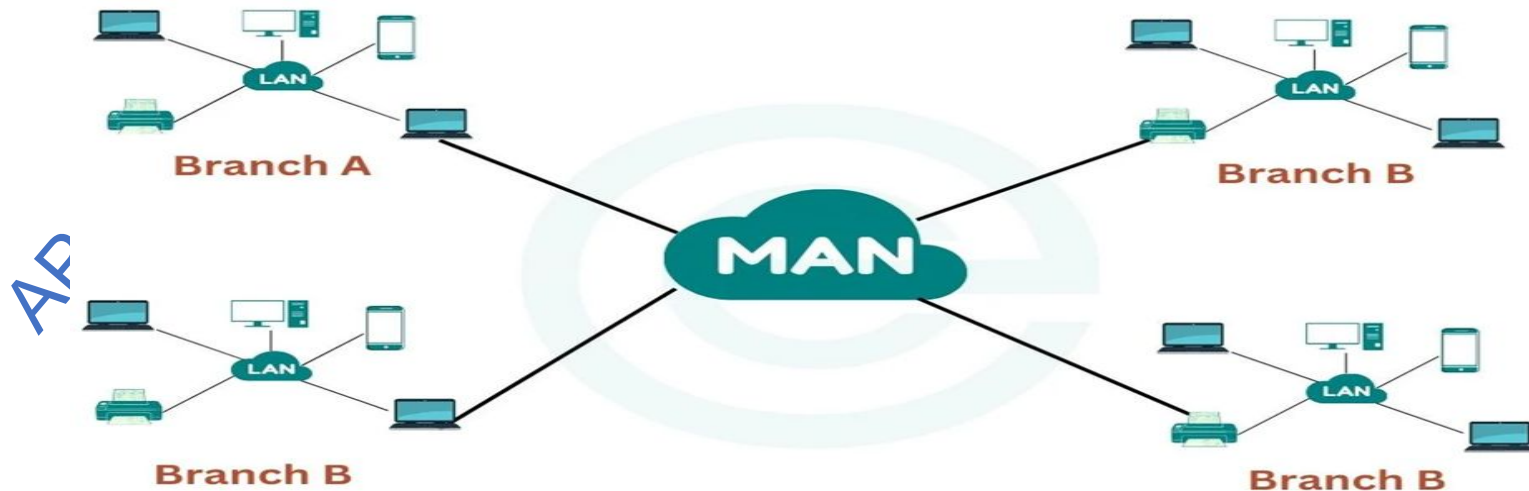
- ❖ It is a type of network that connects multiple local area networks (LANs) within a city or a large campus. (few kilometers to around 50 kilometers)
- ❖ It spans a geographic area larger than a LAN but smaller than a Wide Area Network (WAN).
- ❖ MANs are commonly used by governments, educational institutions, and large organizations to share resources and data across different buildings or departments within the city.
- ❖ For example, a city's public libraries or university campuses may be connected through a MAN.

Advantages:

- ✓ Connects multiple LANs
- ✓ High-speed communication over a city

Disadvantages:

- ✓ Setup cost may be high
- ✓ Needs maintenance
- ✓ Server or cable failure can stop network



1.1 Network Types

4. Wide Area Network (WAN)

- ❖ A Wide Area Network (WAN) is a type of network that spans large geographical areas, such as cities, countries, or even continents.
- ❖ It connects computers and smaller networks (like LANs and MANs) using technologies like telephone lines, satellite links, or fiber-optic cables.
- ❖ WANs allow users in distant locations to communicate and share resources efficiently. Common examples include the Internet and banking networks that connect branches across the globe.

Advantages:

- ✓ Can connect thousands of computers
- ✓ Connects multiple LANs from different places
- ✓ Good for large businesses and government

Disadvantages:

- ✓ Slow speed compared to LAN/MAN
- ✓ Expensive setup
- ✓ Requires complex software and skilled management



1.1 Network Types

Network Type	Coverage Area	Common Cables / Media Used	Examples
PAN	Personal Space (1–10 meters)	- USB Cable- Bluetooth (wireless) - Infrared (wireless)	Phone to smartwatch via Bluetooth Mouse connected via USB
LAN	Within a building or Campus	- Twisted Pair (Cat5e, Cat6) - Fiber Optic Cable - Coaxial Cable	Office computers connected via Ethernet Fiber backbone in a campus
MAN	City or Metropolitan Area	- Fiber Optic Cable - Leased Lines - Microwave Links (wireless)	University campuses across a city Bank branches in the same city
WAN	Country to Global Scale	- Fiber Optic Cable - Satellite Links - Undersea Cables - Microwave Links	Internet, telecom infrastructure Cloud services across continents

1.2 Network Topology

What is topology?

- ❖ Topology is the **geometric arrangement** that shows how all nodes and links are positioned and connected in a network.
- ❖ It refers to the physical layout of devices and cables in a network.
- ❖ It shows how devices (nodes) are physically connected.
- ❖ A topology is formed when multiple links and nodes are arranged in a certain shape.

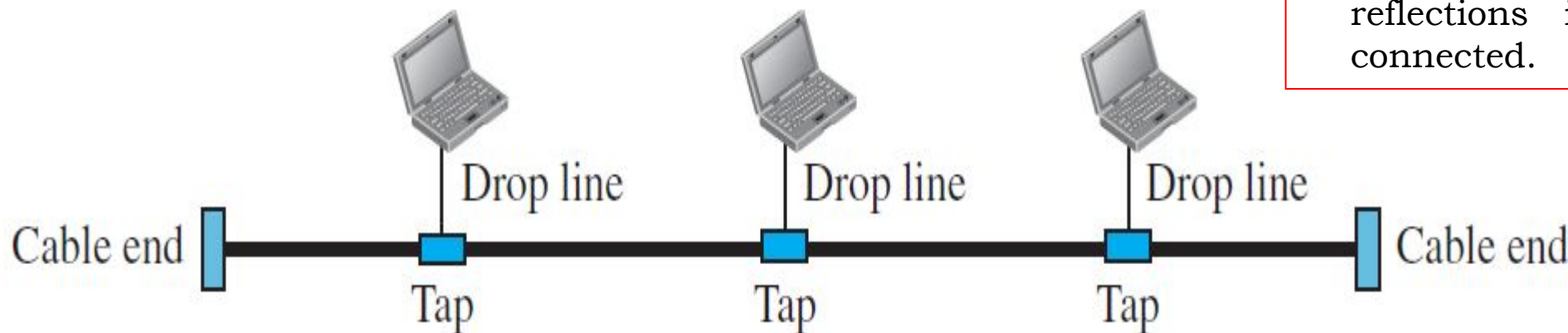
There are four basic types:

1. Bus Topology – All devices share a single communication line (backbone).
2. Star Topology – All devices are connected to a central hub or switch.
3. Ring Topology – Each device is connected to two other devices, forming a closed loop.
4. Mesh Topology – Every device is connected to every other device.
5. Hybrid Topology - It is formed by mixing two or more basic topologies

1.2 Network Topology

1. Bus Topology

- ❖ Bus topology is a multipoint network.
- ❖ It uses one main long cable (backbone) to connect all devices.
- ❖ Devices are attached using:
 - ✓ Drop line – connects a device to the main cable.
 - ✓ Tap – connects drop line to the backbone.
- ❖ Signal starts from one device and travels down the backbone.
- ❖ As it moves, the signal weakens (energy lost as heat).
- ❖ Hence, only a limited number of devices and distance are allowed.



Advantages

- ❑ **Easy to install** – cable runs in a straight line.
- ❑ **Less cabling** – only one main cable is needed.
- ❑ **Cost-effective** – fewer cables compared to star or mesh topology.

Disadvantages

- ❑ **Difficult to add/remove devices** – not flexible.
- ❑ **Fault isolation is hard** – if one point fails, whole network may go down.
- ❑ **Signal quality degrades** – due to reflections if too many devices are connected.

1.2 Network Topology

2. Star Topology

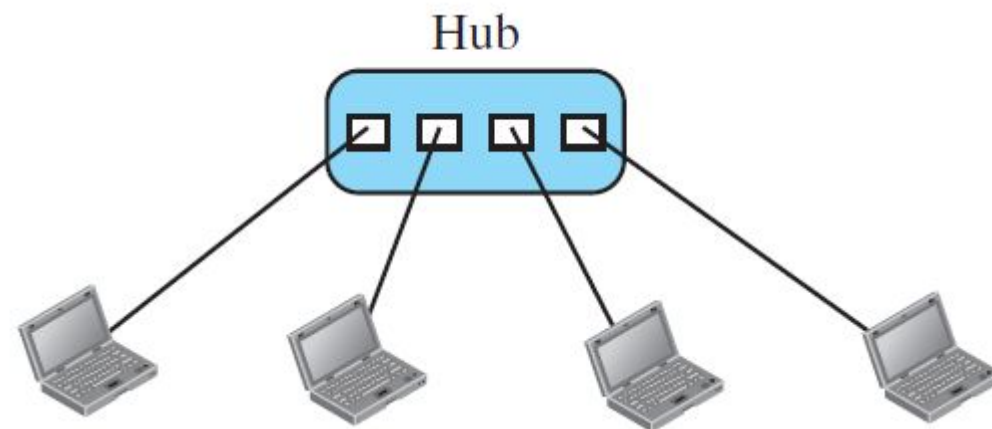
- ❖ A star topology connects all devices to a central hub or switch using point-to-point links.
- ❖ Devices do not connect directly to each other.
- ❖ When one device wants to send data to another:
- ❖ It sends the data to the hub.
- ❖ The hub forwards the data to the correct destination device.
- ❖ Cheaper than mesh topology – each device needs only one cable and one port.
- ❖ Easy to install and reconfigure (add/remove/move devices).
- ❖ Uses less cable than mesh.
- ❖ But more than bus or ring, since every device needs a separate cable to the hub.

Advantages

- ❑ Robust system – If one device or its cable fails, the rest of the network continues working.
- ❑ Easy to troubleshoot – The hub can detect and isolate faults.

Disadvantages

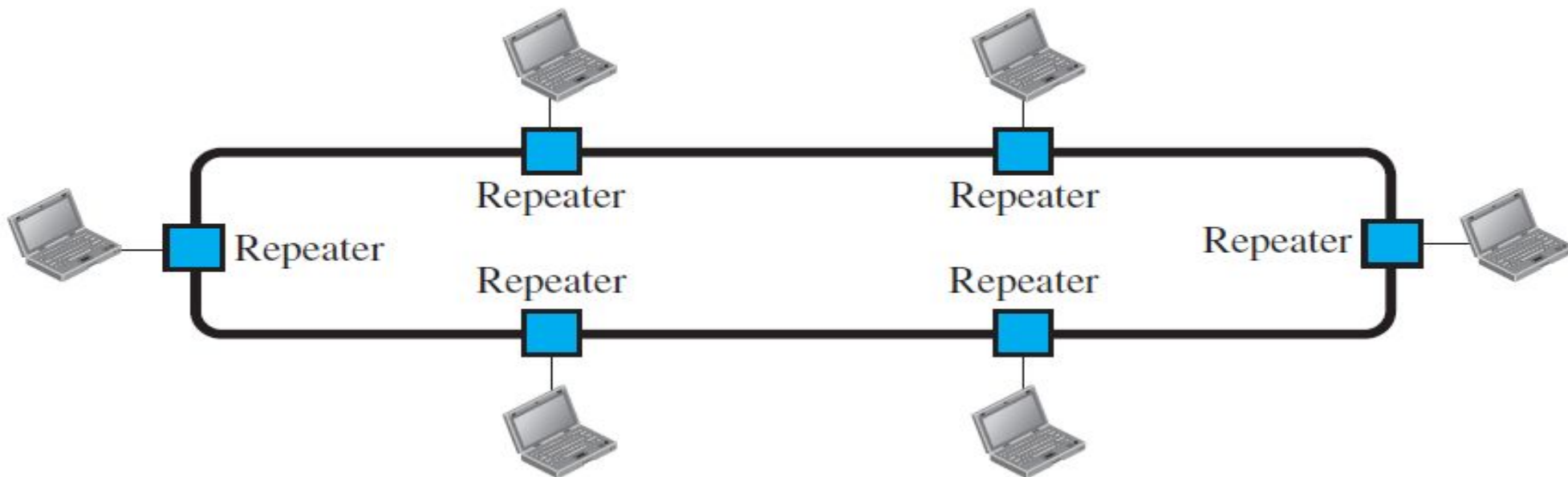
- ❑ Single point of failure – If the hub fails, the entire network stops.
- ❑ Needs more cable than bus or ring for small networks.



1.2 Network Topology

3. Ring Topology

- ❖ In ring topology, each device is connected to exactly two other devices, forming a closed loop or ring.
- ❖ Every connection is point-to-point between neighboring devices.
- ❖ Data travels in one direction (unidirectional) around the ring.
- ❖ The signal passes device to device until it reaches the intended destination.



1.2 Network Topology

3. Ring Topology

- ❖ Each device has a repeater to regenerate and forward the signal to the next device.
- ❖ This ensures the signal remains strong throughout the ring.
- ❖ Ring topology is relatively easy to install and modify.
- ❖ To add or remove a device, only two connections need to be changed (neighbor links).
- ❖ Signals circulate continuously. If a device fails to receive a signal in time, it raises an alarm.
- ❖ This helps in quick fault detection and isolation.

Advantages

- ☐ Simple Installation
- ☐ Equal Access
- ☐ Efficient Performance
- ☐ No Central Hub Needed
- ☐ Fault Isolation

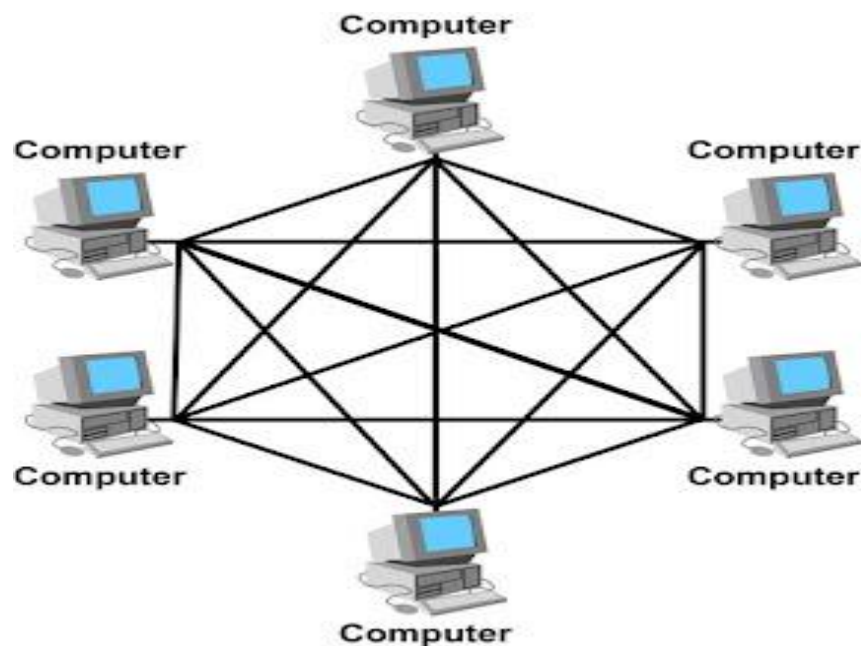
Disadvantages

- ☐ Break Disrupts Entire Network
- ☐ Difficult Troubleshooting
- ☐ Data Delay
- ☐ Unidirectional Flow
- ☐ Slower in Heavy Traffic
- ☐ Cost

1.2 Network Topology

4. Mesh Topology

- ❑ In mesh topology, every device is connected directly to every other device via a dedicated point-to-point link.
- ❑ No link is shared — each connection is exclusively for two devices.
- ❑ For n devices, the number of links = $n(n - 1)/2$ (assuming duplex communication).
- ❑ Each device must have $n - 1$ I/O ports to connect to others.
- ❑ Data flows directly from source to destination without passing through intermediate devices.



$$\begin{aligned}\text{Number of nodes } n &= 6 \\ \text{Number of links} &= n(n-1)/2 \\ &= 6(6-1)/2 \\ &= 6(5)/2 \\ &= 30/2 \\ &= 15\end{aligned}$$

1.2 Network Topology

4. Mesh Topology

- ☐ No data collisions, as links are not shared.
- ☐ If one link fails, it does not affect the entire network.
- ☐ Alternative paths are always available due to multiple connections.
- ☐ Fault detection and isolation are easier due to point-to-point links.
- ☐ Faulty links can be easily identified and bypassed.
- ☐ Dedicated communication paths ensure data is not accessed by other devices.
- ☐ Physical isolation enhances security.
- ☐ Commonly used in critical systems, like telephone networks and data center backbones.

Advantages

- ☐ Dedicated Links
- ☐ Robust & Reliable
- ☐ Privacy and Security
- ☐ Easy Fault Detection
- ☐ Multiple Paths

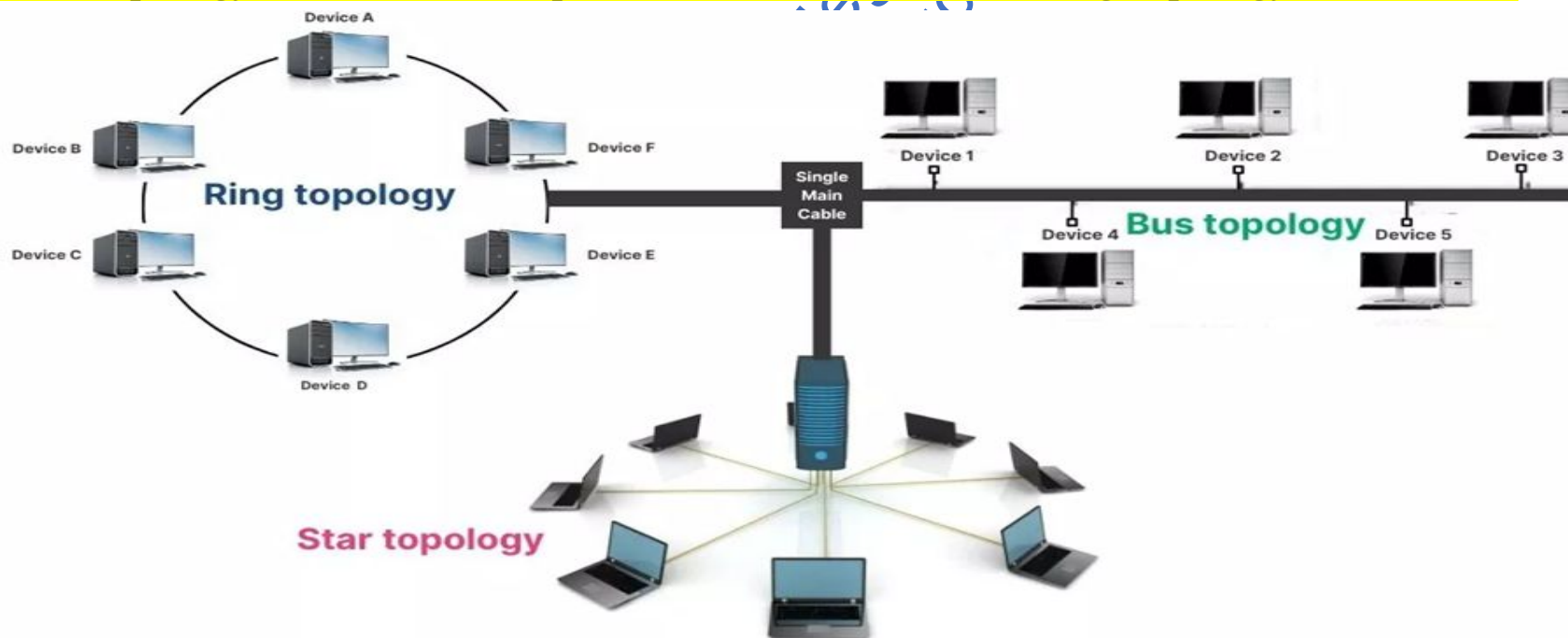
Disadvantages

- ☐ High Cabling Cost
- ☐ Complex Installation
- ☐ Space Constraints
- ☐ Expensive Hardware
- ☐ Scalability Issues

1.2 Network Topology

5. Hybrid Topology

- ❑ A hybrid topology is a combination of two or more different network topologies, such as star, ring, bus, and mesh.
- ❑ It is designed to leverage the strengths of each topology and reduce their weaknesses.
- ❑ Different topologies are interconnected to form a larger, more complex network.
- ❑ **Example:** A star topology used in one department connected to a ring topology in another.



1.2 Network Topology

5. Hybrid Topology

- ☐ Highly customizable based on organizational needs.
- ☐ Suitable for large-scale and complex networks.
- ☐ New nodes or networks can be easily added without affecting the existing structure.
- ☐ Ideal for organizations that plan for future expansion.
- ☐ Large enterprises like banks, universities, or corporate offices often use hybrid topology.

Advantages

- ☐ Flexible Design
- ☐ Scalable
- ☐ Reliable and Efficient
- ☐ Enhanced Performance
- ☐ Better Fault Isolation

Disadvantages

- ☐ Complex Design
- ☐ High Cost
- ☐ Maintenance Challenges
- ☐ Compatibility Issues

1.3 Switching

What is a Switching?

- ❖ Switching is a mechanism that provides communication between different networks or different computer(s) and manages the data flow between the two end points.
- ❖ There are two types of switching techniques –
 1. Circuit Switching
 2. Packet Switching
 3. Message Switching

1. Circuit - Switched:

- ❖ It is a switching method in which a **dedicated physical path** is formed between two points in a network i.e. between the sending and the receiving devices.
- ❖ These dedicated paths are created by a set of switches connected by physical links.
- ❖ Circuit switching is the simplest method of data communication that has a fixed data rate and both the subscribers need to operate at this fixed rate.

1.3 Switching

Phases Of Circuit switching

1. Setup Phase

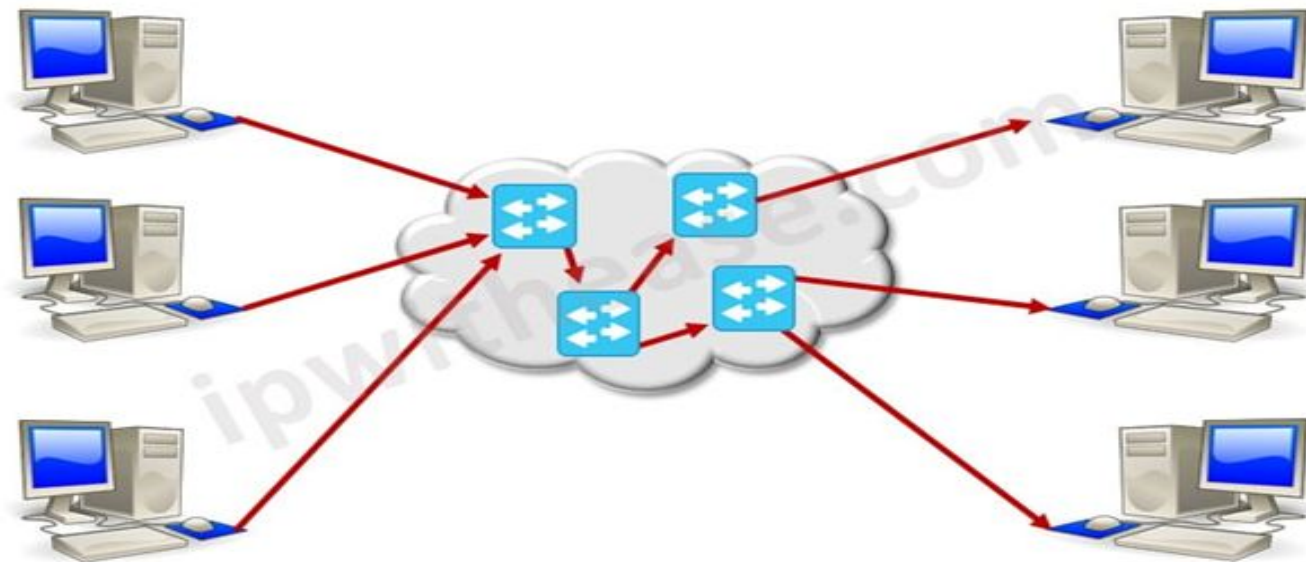
- ❖ A dedicated circuit or path is established between the sender and receiver before the actual data transfer.

2. Data Transfer Phase

- ❖ Data transfer only starts after the setup phase is completed and a physical, dedicated path is established.
- ❖ The data flow is continuous and there may be periods of silence in data transmitting.

3. Disconnect or Teardown Phase

- ❖ When one of the subscribers (either the sender or the receiver) needs to disconnect, a disconnect signal is sent to each switch to release the resource and break/disconnect the connection.



Real-Time Scenario: Traditional Telephone Call

- ❖ Imagine you're using a landline phone to call someone:
- ❖ When you dial the number, the telephone network sets up a dedicated path from your phone to the other person's.
- ❖ This path remains reserved exclusively for your conversation—no one else can use it until the call ends.
- ❖ The voice data flows continuously and in order over that fixed route.
- ❖ Once the call is complete, the circuit is turned down and the resources are freed.

1.3 Switching

Limitations of Circuit Switched

Constraint	Why It Fails in Your Scenario
Dedicated path needed	WhatsApp and Facebook run over the internet, not dedicated circuits
Inefficient resource usage	A continuous path for each chat session would waste bandwidth during idle times
No simultaneous app use	Two parallel circuit-switched connections per app would be complex and impractical
Global scalability issue	Circuit switching doesn't scale well to millions of dynamic internet-based chats

1.3 Switching

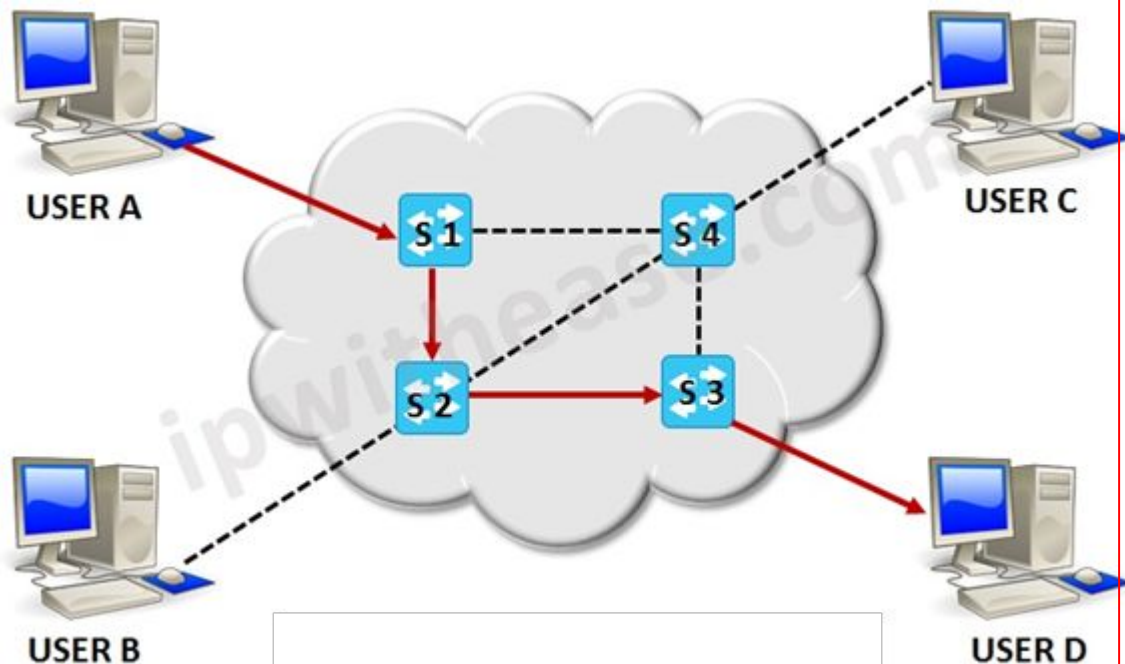
2. Packet switched

- ❖ Packet switching is a switching method in which data is divided into **small units** of variable length called packets.
- ❖ Each packet contains part of the **data** along with information like the **destination address**.
- ❖ These packets are transmitted over the network, possibly taking **different paths**, and are reassembled at the destination to form the original message or file.
- ❖ Thus, Packet Switching method uses **Store and Forward technique**.
- ❖ Each node will store the packet and then forward the packets to the next host destination.
- ❖ There are two types of packet switching :
 - i. Virtual Circuit Packet Switching
 - ii. Datagram Packet Switching

1.3 Switching

i. Virtual Circuit(VC) Packet Switching

- ❖ virtual circuit or logical path connection is done between sender and receiver.
- ❖ A predefined route is established, and all packets follow this path using a **unique Virtual Circuit ID** assigned to each router or switch to identify the logical connection.
- ❖ It also has the same three-phase protocol used in circuit switching – Setup Phase, Data Transfer Phase and Tear down Phase.



Step 1: Connection Request:

- ❖ User A initiates communication with User D.
- ❖ A request is sent to establish a virtual circuit through the network.

Step 2: Path Selection

- ❖ The network determines a predefined path:
- ❖ User A → Switch S1 → Switch S2 → Switch S3 → User D.

Step 3: Virtual Circuit ID Assignment

- ❖ Each switch on the path (S1, S2, S3) assigns a unique Virtual Circuit ID (VCI).
- ❖ This ID helps each switch identify and forward packets related to that specific connection.

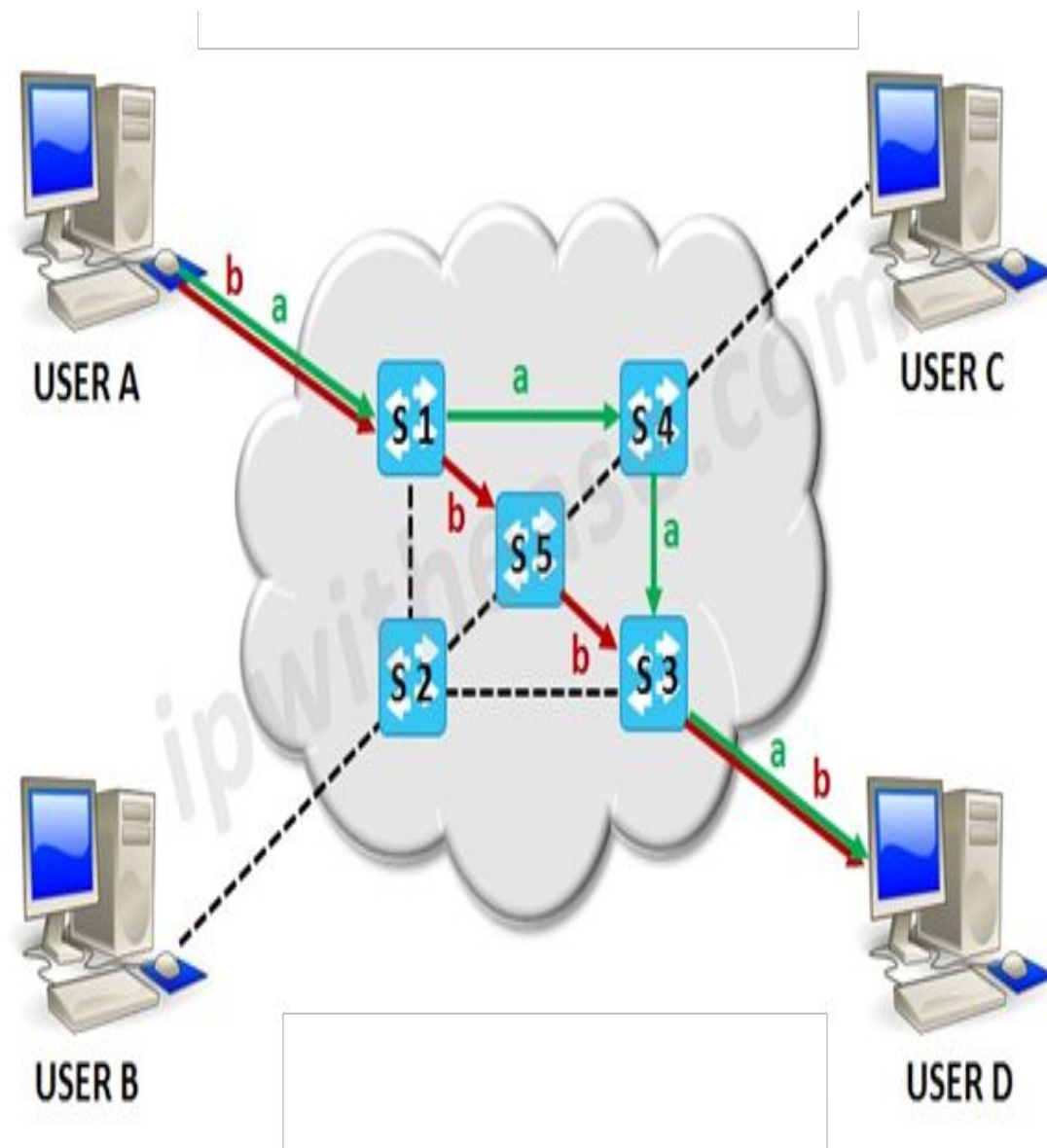
Step 4: Path Confirmation

- ❖ Once the route is set and all VCIs are assigned, a confirmation is sent back to User A, indicating that the path is ready.

1.3 Switching

ii. Datagram Packet Switching

- ❖ The original message is divided into small packets called datagrams.
- ❖ Each datagram is equipped with complete addressing information, including:
 - ☐ Source Address
 - ☐ Destination Address
 - ☐ Sequence Number (optional)
- ❖ Each datagram is sent independently through the network.
- ❖ Routers make routing decisions per packet, possibly choosing different paths for different packets based on:
 - ☐ Network traffic
 - ☐ Link availability
 - ☐ Routing algorithms

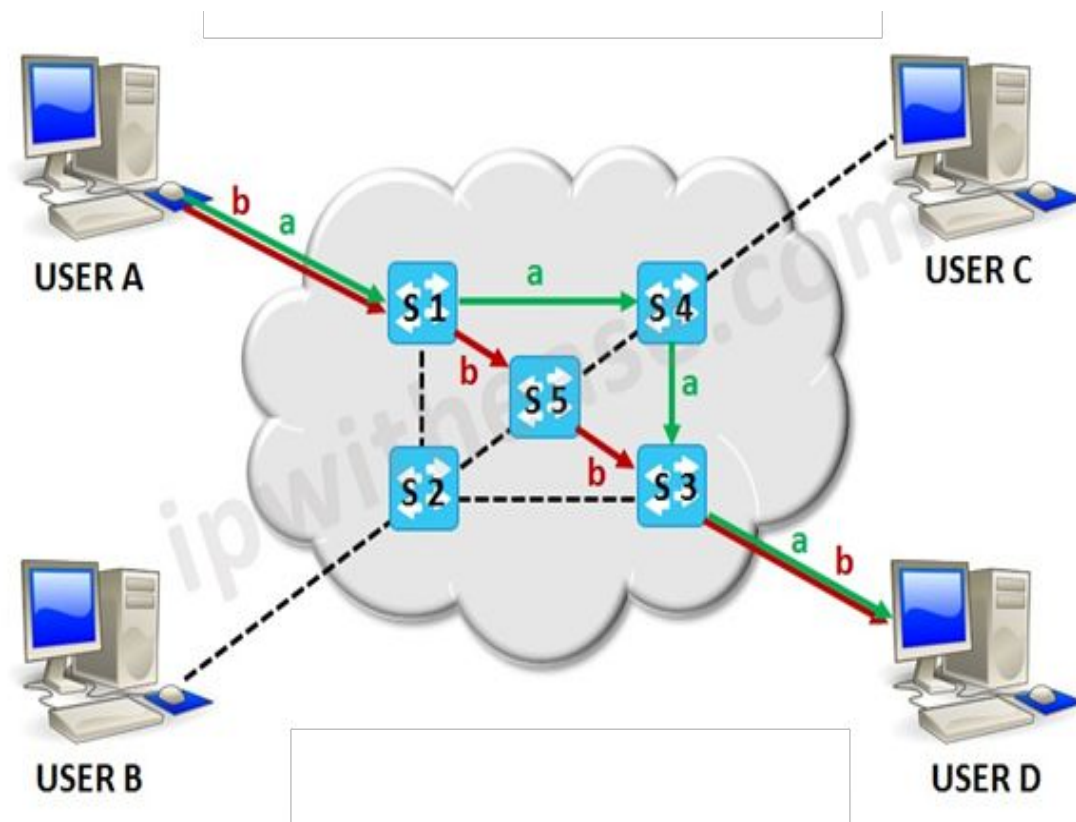


1.3 Switching

ii. Datagram Packet Switching

- ❖ Unlike circuit switching or virtual circuits, there is no predefined path.
- ❖ Packets may arrive at the destination:
 - ☐ Out of order
 - ☐ At different times
 - ☐ Or may even be lost due to congestion
- ❖ The destination host receives the datagrams in whatever order they arrive.
- ❖ It may temporarily store the packets in a buffer.
- ❖ An end-system protocol like TCP reorders the packets and requests retransmission of any lost packets.
- ❖ This ensures reliable and correct delivery to the application.

[Packet Switching.gif](#)



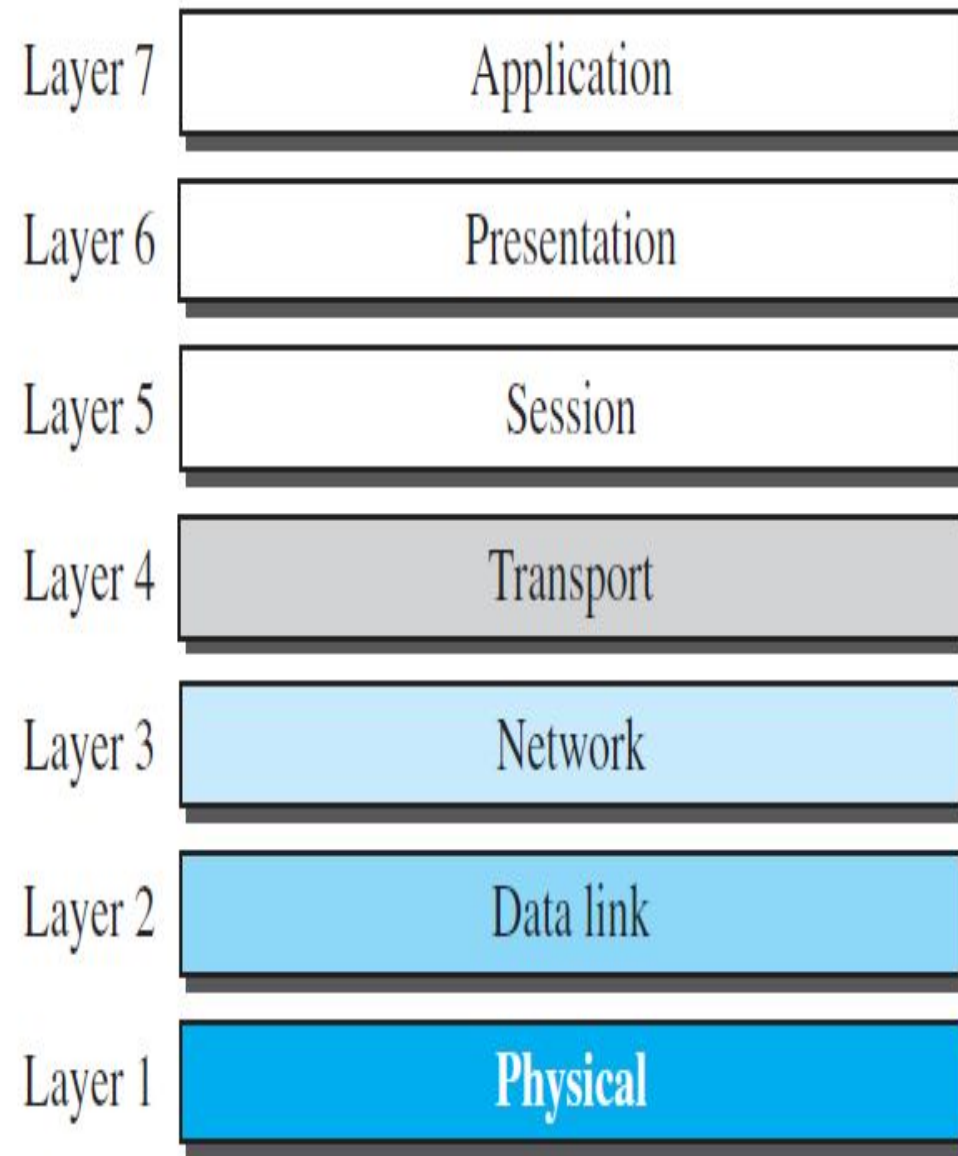
Datagram switching does not require:

- ☐ Connection setup
- ☐ Resource reservation
- ☐ Connection teardown

1.4 OSI Layered Architecture

Layered Architecture

- ❖ A **protocol** is a set of rules that provides communication between sender, receiver and any intermediate devices to ensure proper data exchange.
- ❖ In **simple communication**, only one protocol may be needed.
- ❖ In **complex communication**, tasks are split into layers, each layer handling a part of the process using its own protocol (this is called protocol layering).
- ❖ **Layering** divides a complex task into smaller, manageable units.
- ❖ Each layer performs a **dedicated function** (e.g., encryption, transmission, reception), making the system easier to design and manage.
- ❖ A layer **provides services** to the layer above and **uses services** from the layer below.
- ❖ The **internal working** (implementation) of each layer doesn't need to be known—only the input/output matters.



1.4 OSI Layered Architecture

- ❖ ISO developed a networking standard called the **Open Systems Interconnection (OSI) model**.
- ❖ The model was first introduced in the late 1970s.
- ❖ Definition:
An **open system** includes a set of **protocols** enabling communication between **any two systems**, regardless of their internal **architecture**.
- ❖ The OSI model is not a protocol itself.
- ❖ It is a conceptual framework used for understanding and designing network systems.

Purpose of the OSI Model

- ❖ To enable interoperability between different computer systems.
- ❖ It allows communication without changing the existing hardware or software logic.
- ❖ The goal is to provide a universal framework for designing network architectures.

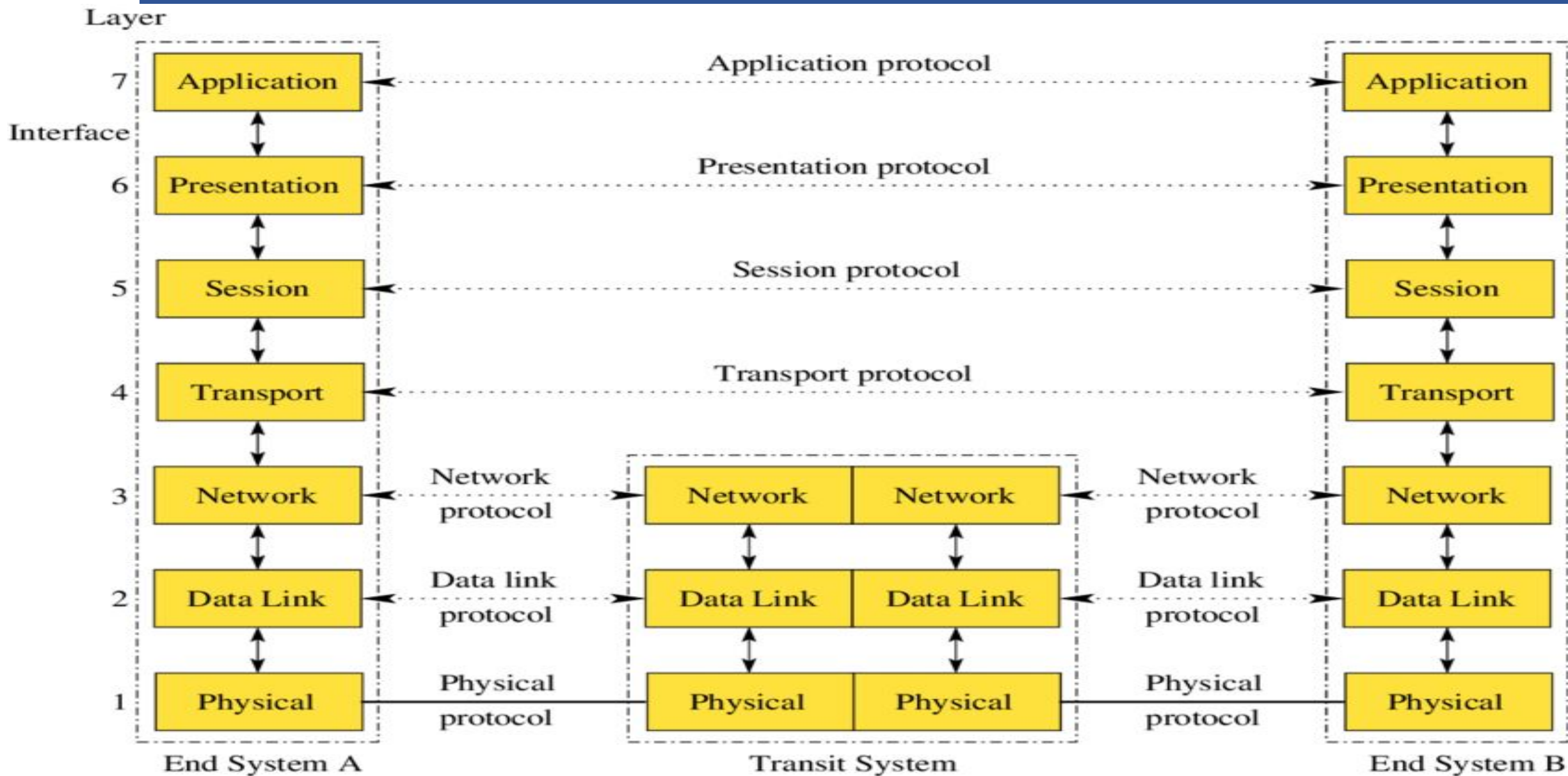
1.4 OSI Layered Architecture

Aim of OSI Model

- ❖ To create a flexible, robust, and interoperable architecture for networks.
- ❖ It was designed to serve as the foundation for creating a comprehensive set of OSI-based network protocols.
- ❖ The OSI model is built on a seven-layer structure.
- ❖ Each layer is independent but interrelated.
- ❖ These layers define specific tasks in the process of data transmission across a network.

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1.4 OSI Layered Architecture



1.4 OSI Layered Architecture

Application Layer

- ❖ The Application Layer is the topmost layer of the OSI model. It provides services directly to the end user or to user applications.
- ❖ This layer enables interaction with software applications that implement a communication component.

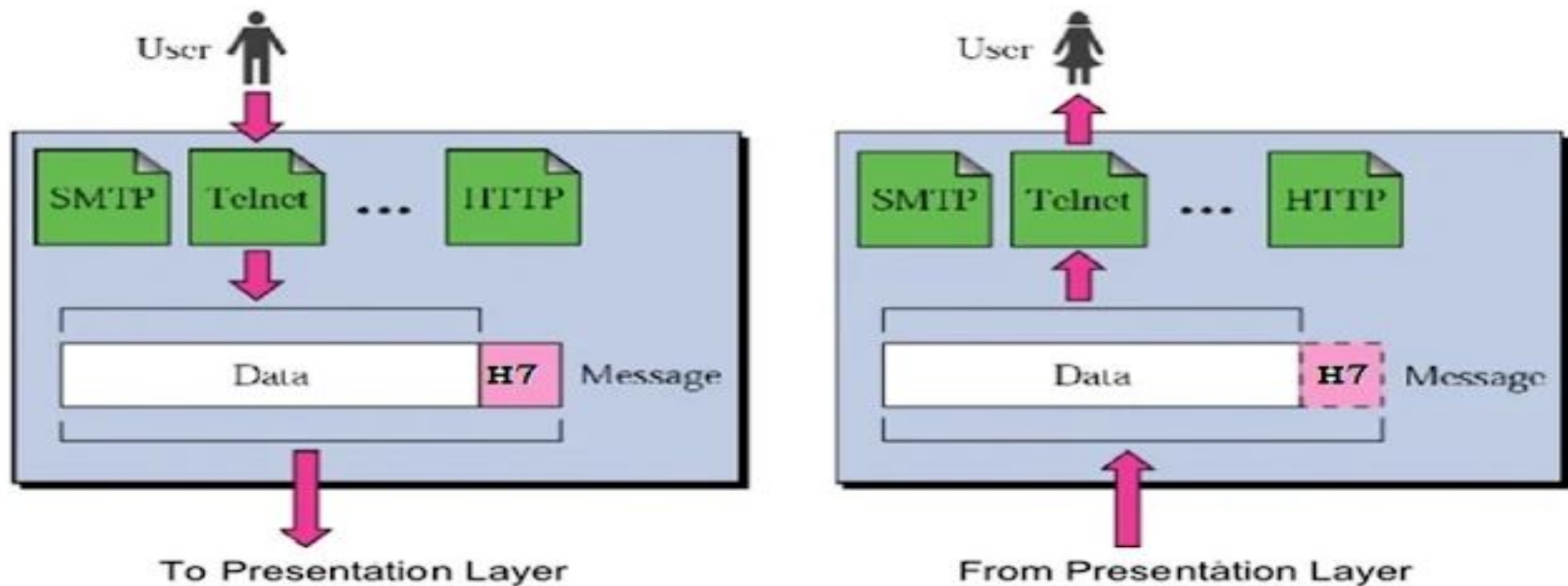
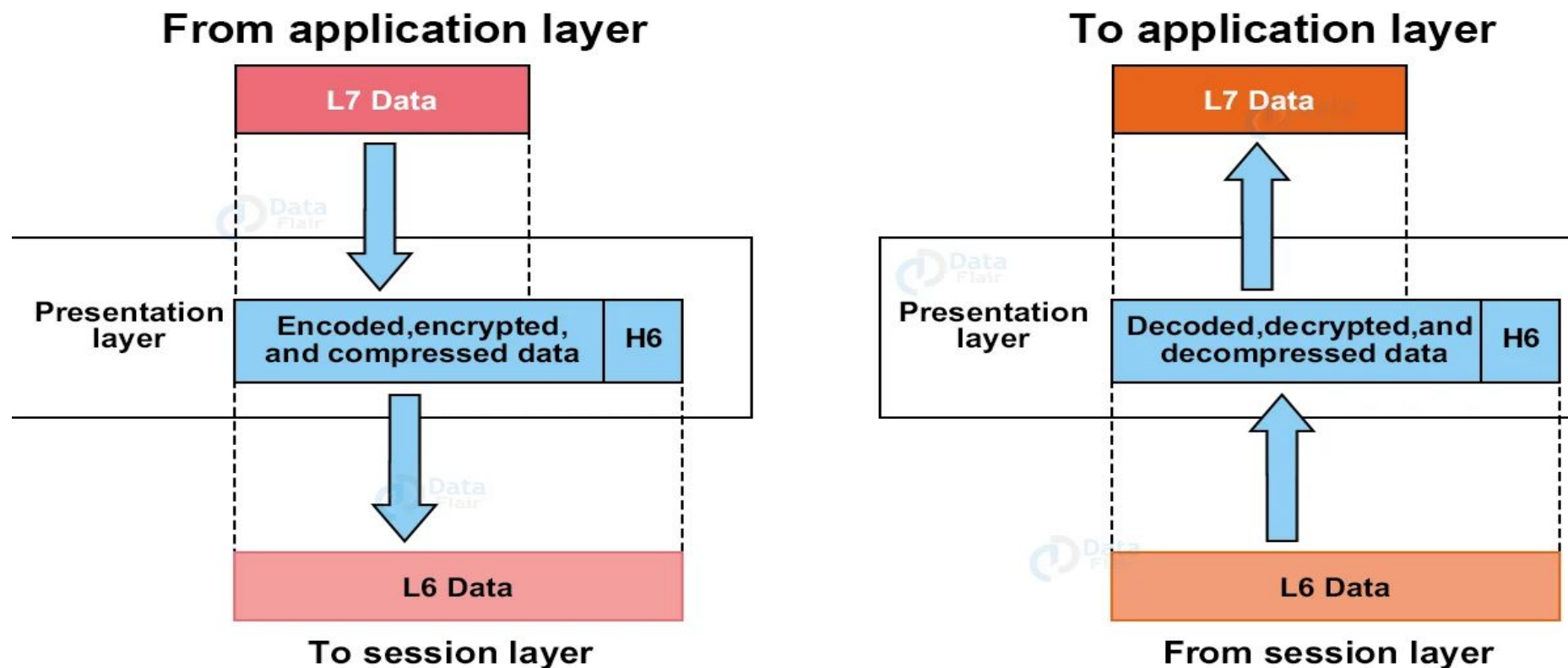


Fig: Application Layer

1.4 OSI Layered Architecture

Presentation Layer

- ❖ **The Presentation Layer** is responsible for managing the **syntax and semantics** of the data exchanged between two systems.
- ❖ It ensures that the data transmitted by the sender is **properly formatted, encoded, and understandable** by the receiver, regardless of differences in data representation.



1.4 OSI Layered Architecture

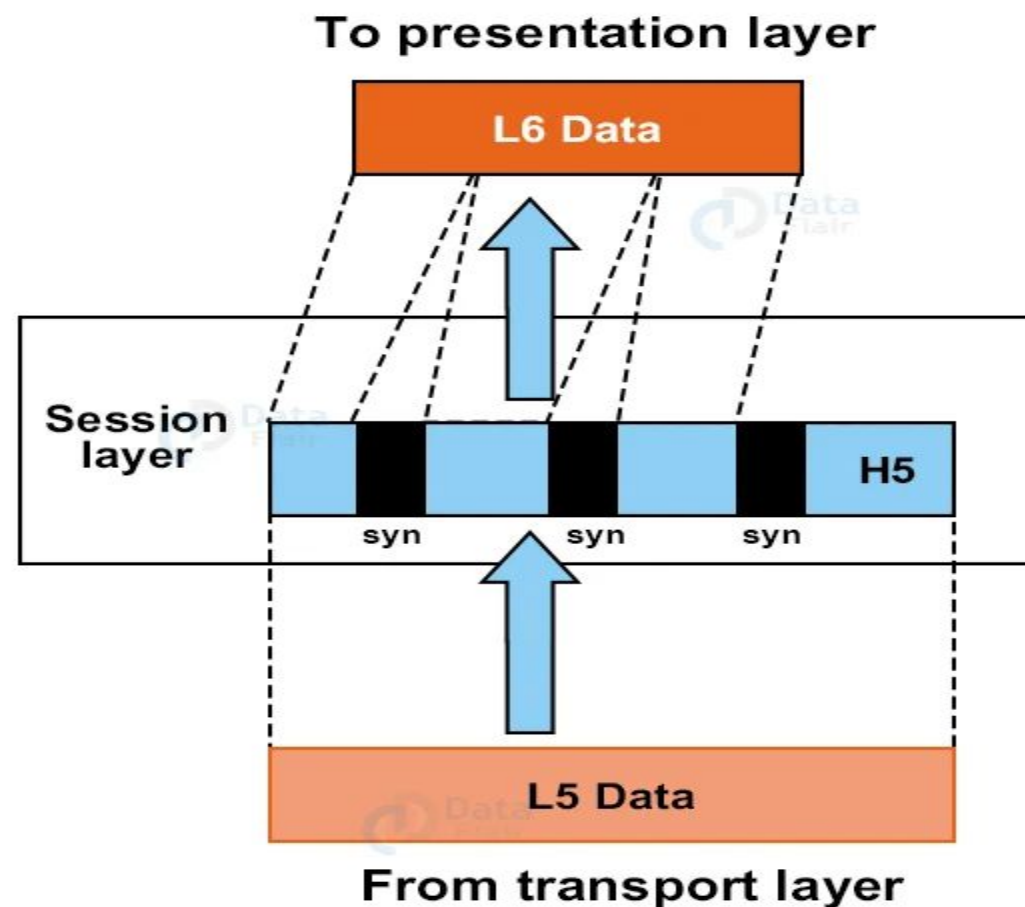
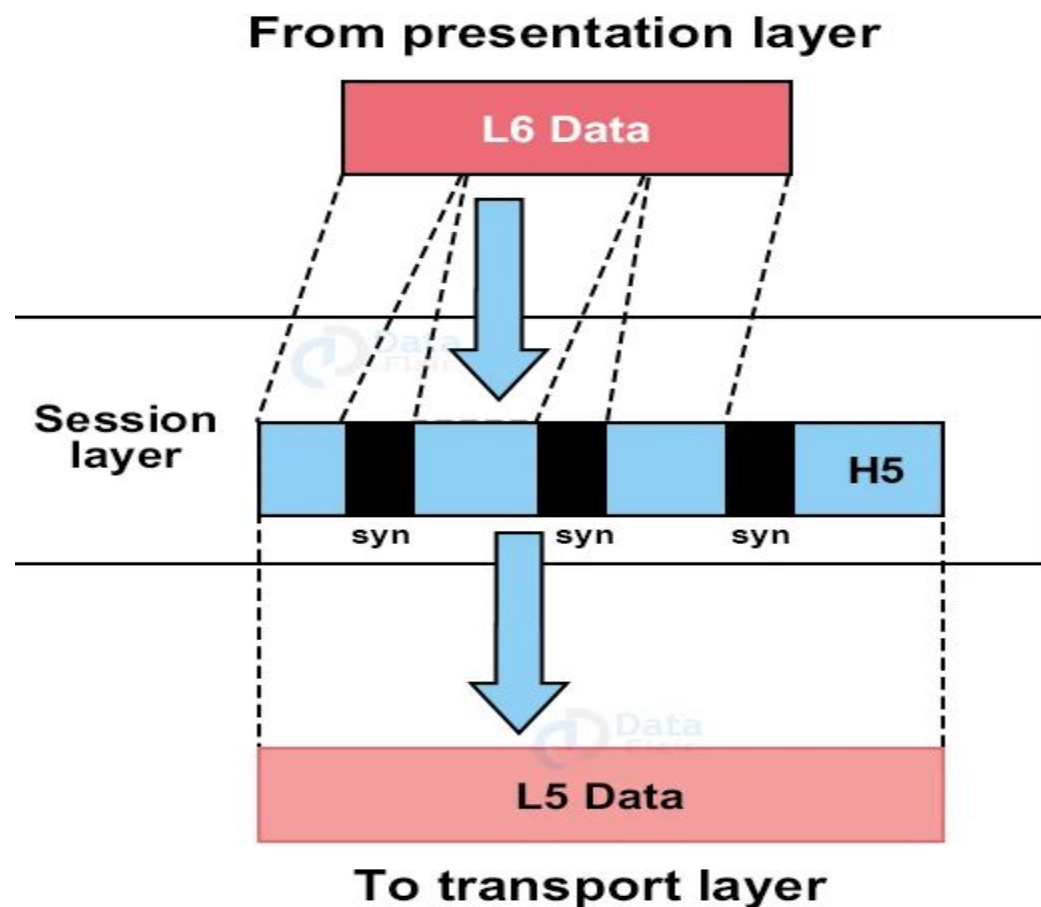
Duties of Presentation layer

Duty	Description
Translation	❖ Converts data between application-specific formats and network standard formats. ❖ Resolves differences in character encoding (e.g., ASCII, EBCDIC). ❖ Handles bit-order translation (e.g., little-endian vs big-endian). ❖ Ensures all systems understand the structure and format of data (e.g., whether 7 or 8 bits represent a character).
Compression	❖ Reduces the number of bits needed to represent the data. ❖ Improves transmission speed and reduces bandwidth usage. ❖ Useful for multimedia files like images, audio, and video.
Encryption and Decryption	❖ Encrypts (scrambles) the data into an unreadable form at the sender's end to ensure data confidentiality. ❖ Decrypts the data back into readable form at the receiver's end. ❖ Provides data security during transmission over the network.

1.4 OSI Layered Architecture

Session Layer

- ❖ The Session Layer is responsible for managing and controlling dialogues (sessions) between two systems.
- ❖ It ensures that communication between applications is properly established, maintained, and terminated.



1.4 OSI Layered Architecture

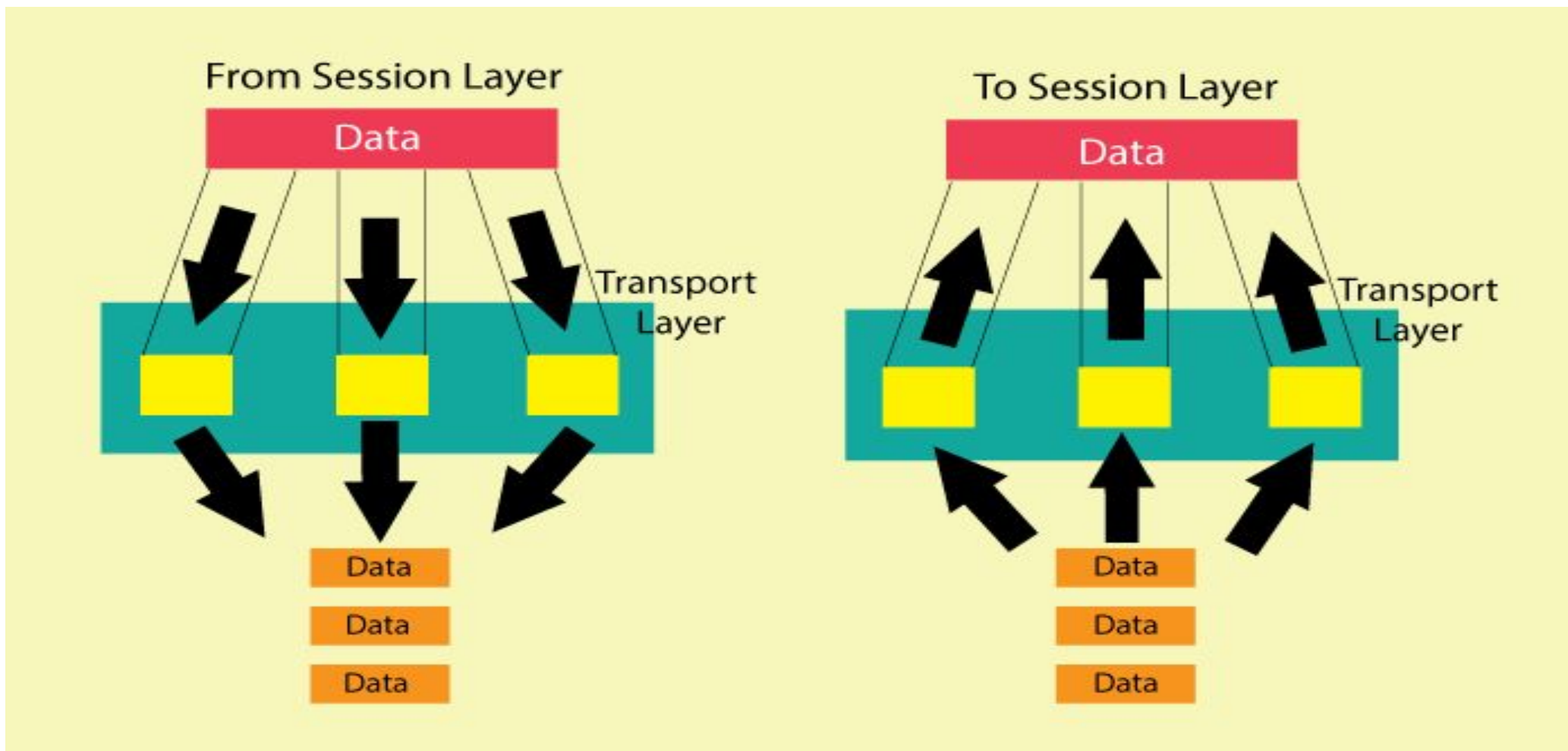
Duties of Session layer

Duty	Description
Session Establishment	❖ Starts the communication between two systems. ❖ Performs authentication to verify the identity of users or systems before allowing communication.
Dialog Control	❖ Controls who can talk and when in a conversation.
Synchronization	❖ Places checkpoints during data transfer. ❖ If the session gets interrupted, it can resume from the last checkpoint, instead of starting over.
Session Termination	❖ Ends the session after communication is complete. ❖ Ensures both systems release resources properly.

1.4 OSI Layered Architecture

Transport Layer

- ❖ The Transport Layer is responsible for process-to-process communication—that is, delivering a complete message from the source process to the destination process across a network.



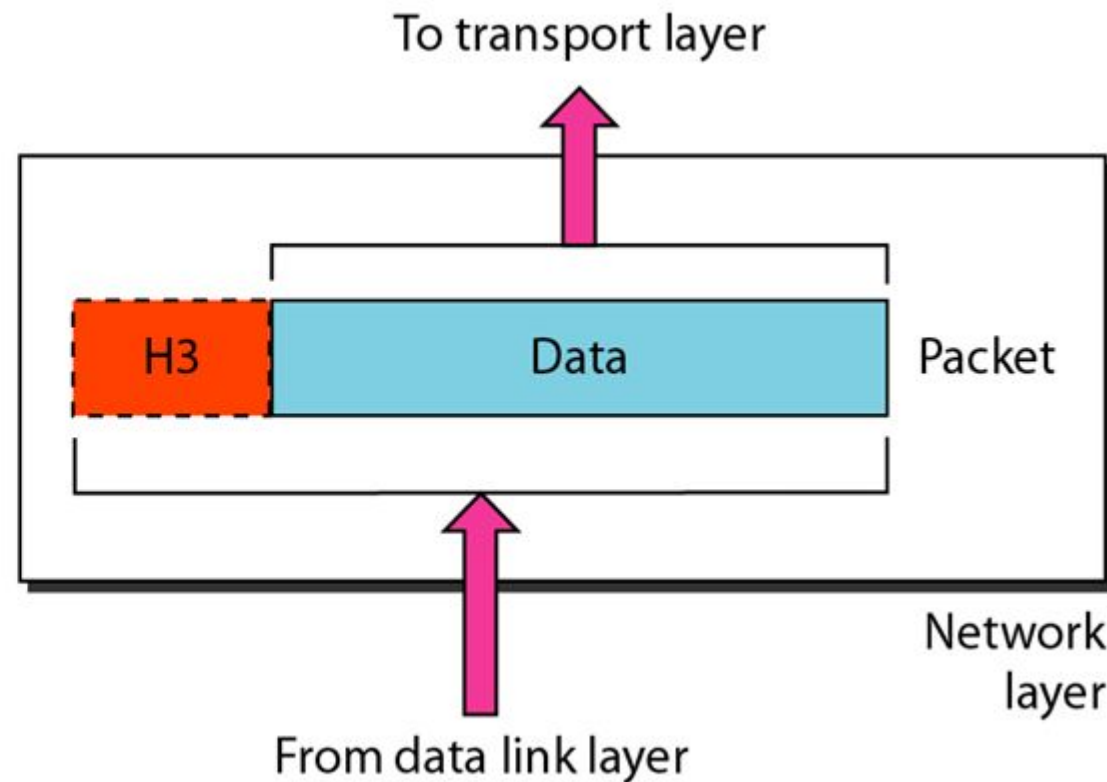
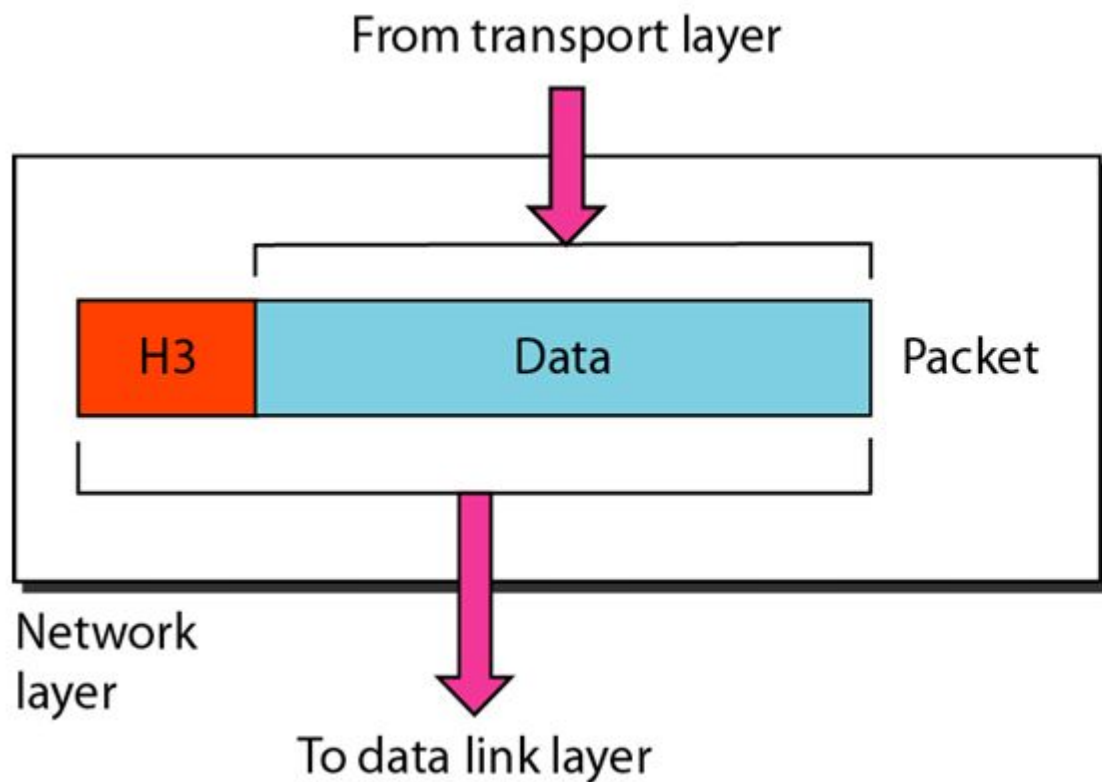
1.4 OSI Layered Architecture

Duties of Transport layer

Duty	Description
Reliable Delivery	❖ Ensures that the entire message is delivered from source to destination in the correct order and without duplication or loss.
Segmentation and Reassembly	❖ Segmentation: Breaks large messages into smaller segments at the sender. ❖ Each segment is numbered to maintain order. ❖ Reassembly: At the receiving end, segments are reassembled into the original message.
Error Control	❖ Ensures error-free delivery by detecting corrupted segments. ❖ If errors are found, retransmission is requested.
Flow Control	❖ Prevents the sender from Sending too much data the receiver by managing data transmission rate.
Connection Control	❖ May be connection-oriented (TCP) or connectionless (UDP). ❖ TCP establishes a virtual connection before data transfer; UDP sends without setup

1.4 OSI Layered Architecture

Network Layer



1.4 OSI Layered Architecture

Network Layer

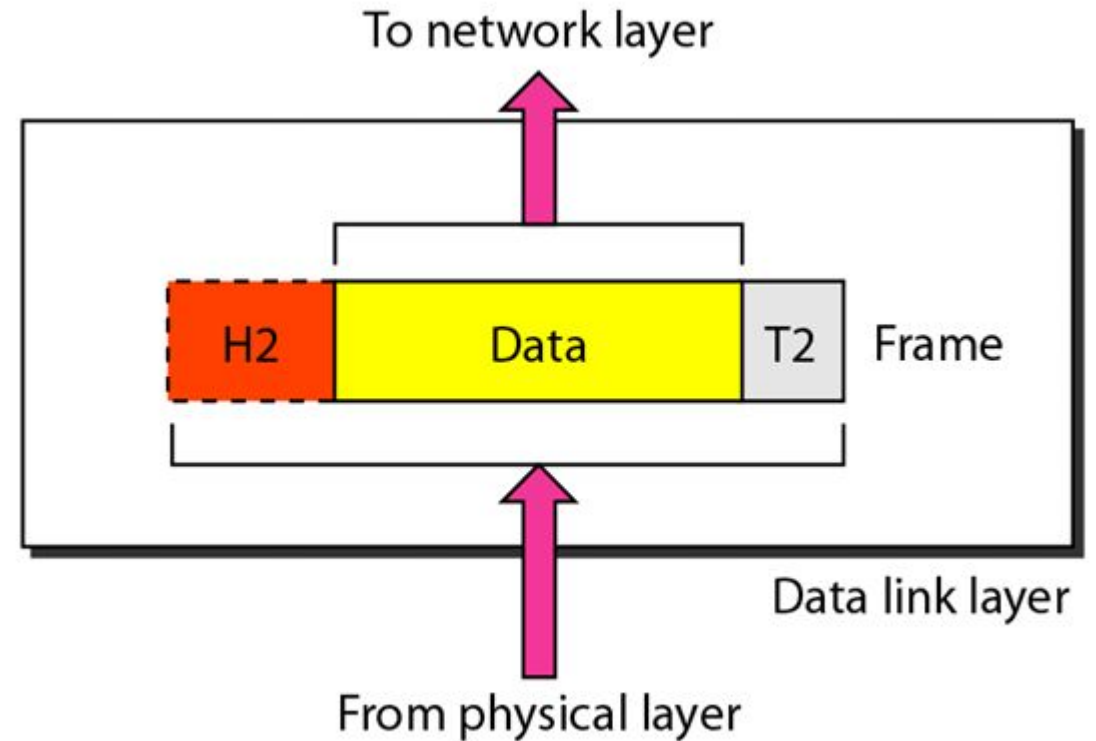
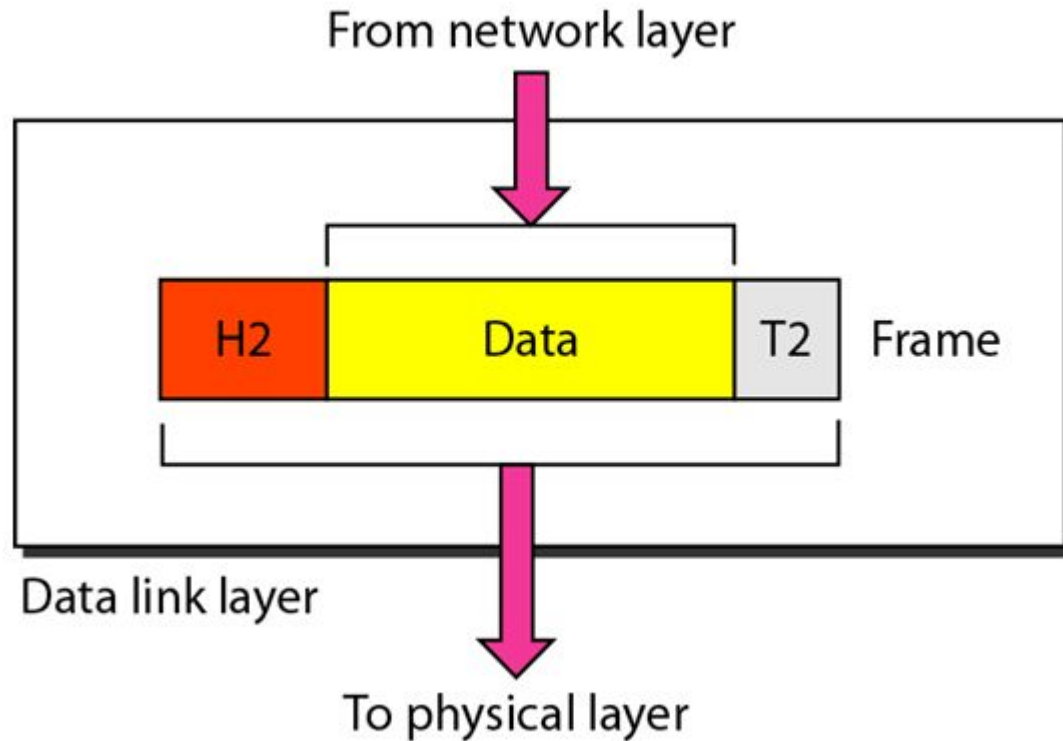
- ❖ The **Network Layer** is mainly responsible for **delivering data packets** from one **host (computer)** to another, even if they are on different networks.
- ❖ This type of communication is called **host-to-host communication**.
- ❖ A packet is a small unit of data. When we send a big file or message, it gets split into smaller packets before transmission.

Duties of Network layer

Duty	Description
Host-to-Host Communication	End-to-end delivery between source and destination
Logical Addressing	Assigning and using IP addresses
Routing	Selecting best path using routers
Packet Forwarding	Moving packets hop-by-hop
Error Reporting	Sending error/status messages (via ICMP)
Multicast/Broadcast Handling	Supporting one-to-many or all-device communication

1.4 OSI Layered Architecture

Datalink Layer



1.4 OSI Layered Architecture

Datalink Layer

- ❖ Responsible for delivery of data between two systems on the same network
- ❖ It takes a packet from the Network Layer and encapsulates it into manageable units called **frames** for transmission over a physical link.
- ❖ **Framing** is the process by which the Data-Link Layer takes a packet (IP datagram) received from the Network Layer and encapsulates it into a frame by adding a header and trailer that contain link-layer-specific information for local delivery.
- ❖ A **frame** is a structured **data unit** used at the **Data-Link Layer** of the OSI model.

A Frame includes:

❑ Header

- ❖ Contains source and destination MAC addresses
- ❖ May also include frame type, control info

❑ Payload

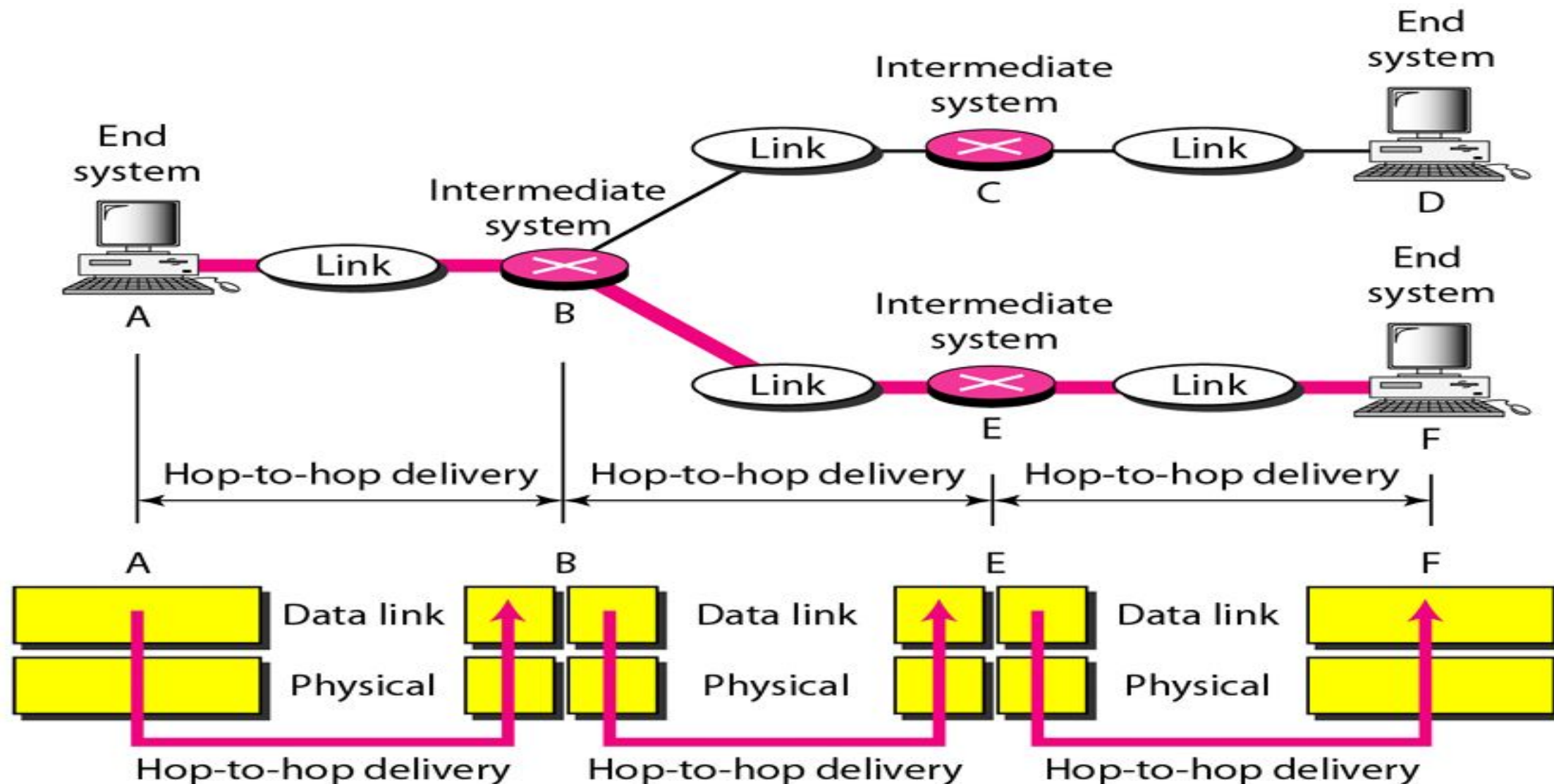
- ❖ The actual data from the Network Layer (e.g., an IP datagram)

❑ Trailer

- ❖ Used for error detection, such as a CRC (Cyclic Redundancy Check)

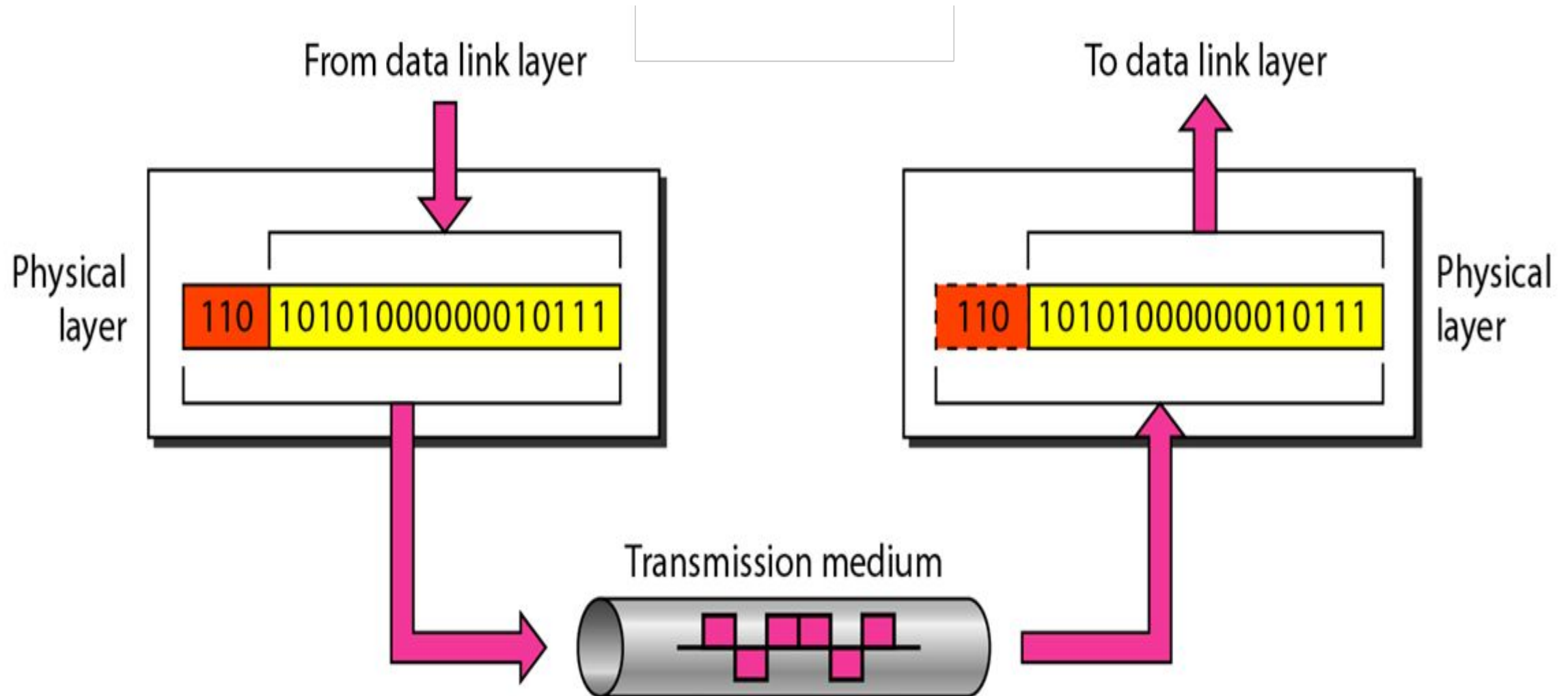
1.4 OSI Layered Architecture

Hop to Hop Delivery



1.4 OSI Layered Architecture

Physical Layer



1.4 OSI Layered Architecture

Physical Layer

- ❖ The Physical Layer is responsible for transmitting raw bits (0s and 1s) over a physical communication channel.
- ❖ It is the lowest layer of the OSI model.
- ❖ It deals with the mechanical, electrical, functional, and procedural characteristics to activate, maintain, and deactivate physical links.

Working of the Physical Layer

- ❖ **Input:** Receives data from the Data Link Layer (in the form of frames).
- ❖ **Process:** Converts frames into a bit stream suitable for the transmission medium.
- ❖ **Transmission:** Sends these bits through a transmission medium (e.g., cable or fiber).
- ❖ **Reception:** At the receiving end, it accepts the bit stream from the medium.
- ❖ **Output:** Passes the received bits to the Data Link Layer of the receiver.

1.4 OSI Layered Architecture

Duties of the Physical Layer

Duty	Description	Examples
1. Characteristics of Media	Specifies type of transmission medium	Twisted pair, coaxial cable, fiber optic
2. Encoding	Converts bit stream (0s and 1s) into signals (electrical, optical, or radio)	Depends on the transmission medium
	Ensures correct representation for the medium	
3. Transmission Rate	Specifies number of bits per second (bps)	Higher bps = faster communication
	Defines bit duration affecting speed and efficiency	
4. Transmission Mode	Determines direction of data flow between devices	See modes below
• Simplex	One-way communication	Keyboard to CPU
• Half-Duplex	Two-way, one direction at a time	Walkie-talkies
• Full-Duplex	Two-way, both directions simultaneously	Telephone calls

Imagine Sending Books to a Friend in Another City – At Sender Side

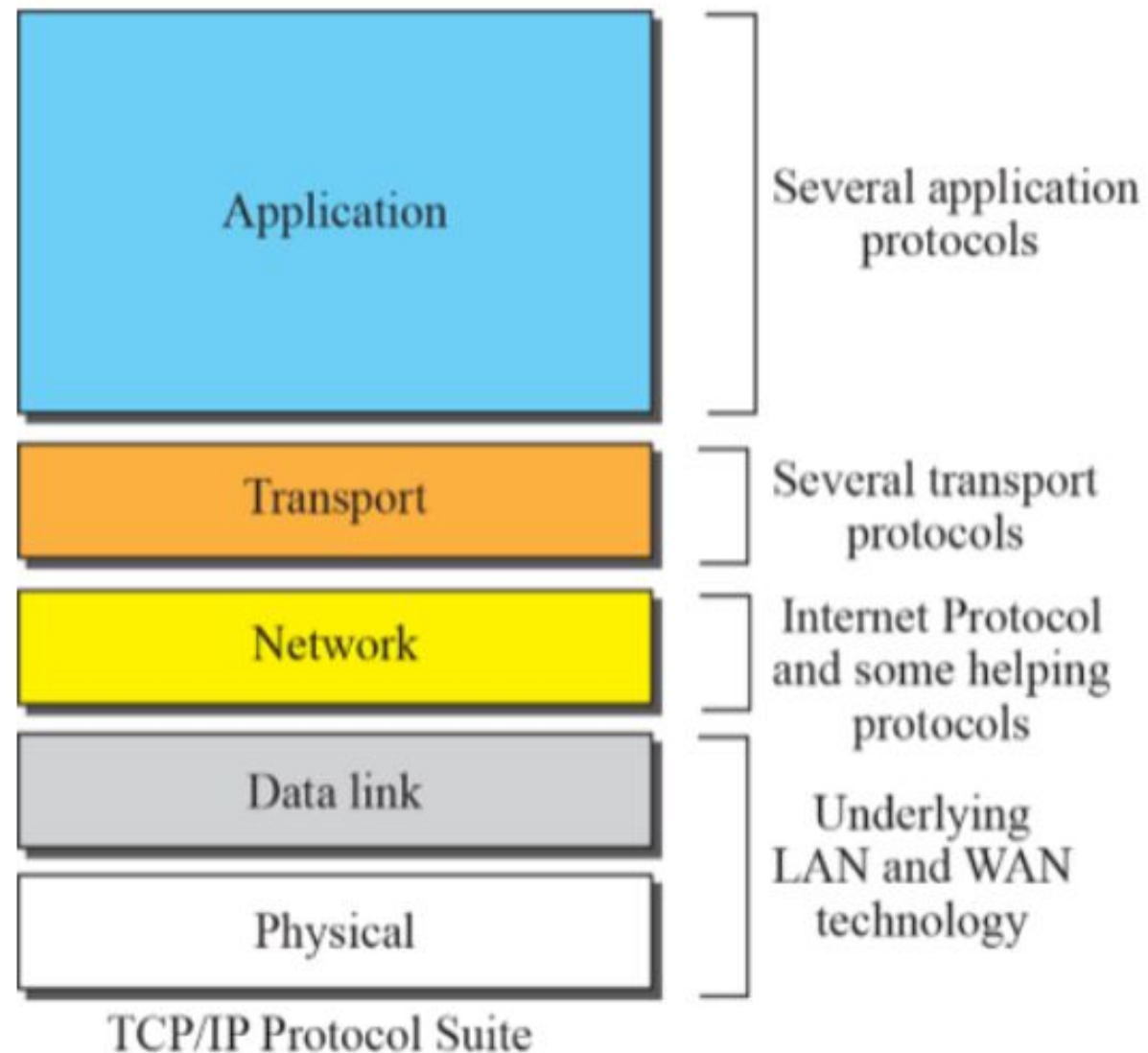
OSI Layer	Book Analogy	Networking Action
7. Application	You decide to send 10 books, write a note for your friend about them.	App creates data (webpage request, email, file transfer).
6. Presentation	You translate your note into a language your friend understands and package it neatly.	Data is encoded, compressed, encrypted if needed (e.g., HTTPS encryption).
5. Session	You call your friend to agree on when and how to send the books.	Session is established and managed between devices (synchronization, checkpoints).
4. Transport (TCP)	You split the books into numbered boxes, record them in a manifest, and agree with your friend to confirm each one’s arrival.	TCP segments data, assigns sequence numbers, ensures reliability, starts 3-way handshake.
3. Network (IP)	You write your friend’s postal address and your return address on each box, then decide the best route across cities to deliver them.	IP adds source & destination IP addresses, uses routing tables to determine the best end-to-end path through multiple networks.
2. Data Link	At each city’s local post office, staff attach local delivery tags to move the box to the next city or neighborhood.	Ethernet/Wi-Fi adds MAC addresses for local hop-to-hop delivery, forwards frames to the next directly connected device, and checks for link-level errors.
1. Physical	Trucks, planes, and conveyors move the boxes physically to your friend’s city.	Bits are sent as electrical signals, light pulses, or radio waves over cables, fiber, or wireless.

Imagine Sending Books to a Friend in Another City – At Receiver Side

OSI Layer	Book Analogy	Networking Action
1. Physical	The delivery truck, plane, or conveyor brings the boxes physically to your friend’s local post office.	Electrical signals, light pulses, or radio waves are received over cables, fiber, or wireless.
2. Data Link	The local post office reads the delivery tag for that neighborhood and forwards the box to your friend’s street.	MAC addresses are checked for local delivery; Ethernet/Wi-Fi frames are processed and link-level errors corrected.
3. Network (IP)	The building’s mailroom checks the postal address, confirms it belongs to your friend, and delivers it to their apartment.	IP verifies the destination IP address, removes the IP header, and passes the segment to TCP. Routing is complete at this point.
4. Transport (TCP)	Your friend opens each box, reorders the books (1–10), requests resends for any missing ones, and confirms arrival of each.	TCP reorders segments, ensures all are received, requests retransmissions if needed, and acknowledges data.
5. Session	You and your friend keep track of this exchange as part of the same book-sending “session.”	Maintains the session state between applications.
6. Presentation	Your friend reads your note in the correct language and format.	Data is decrypted, decompressed, and converted to the correct format.
7. Application	Your friend enjoys the books and understands your message.	The application receives complete data (e.g., web browser displays the webpage).

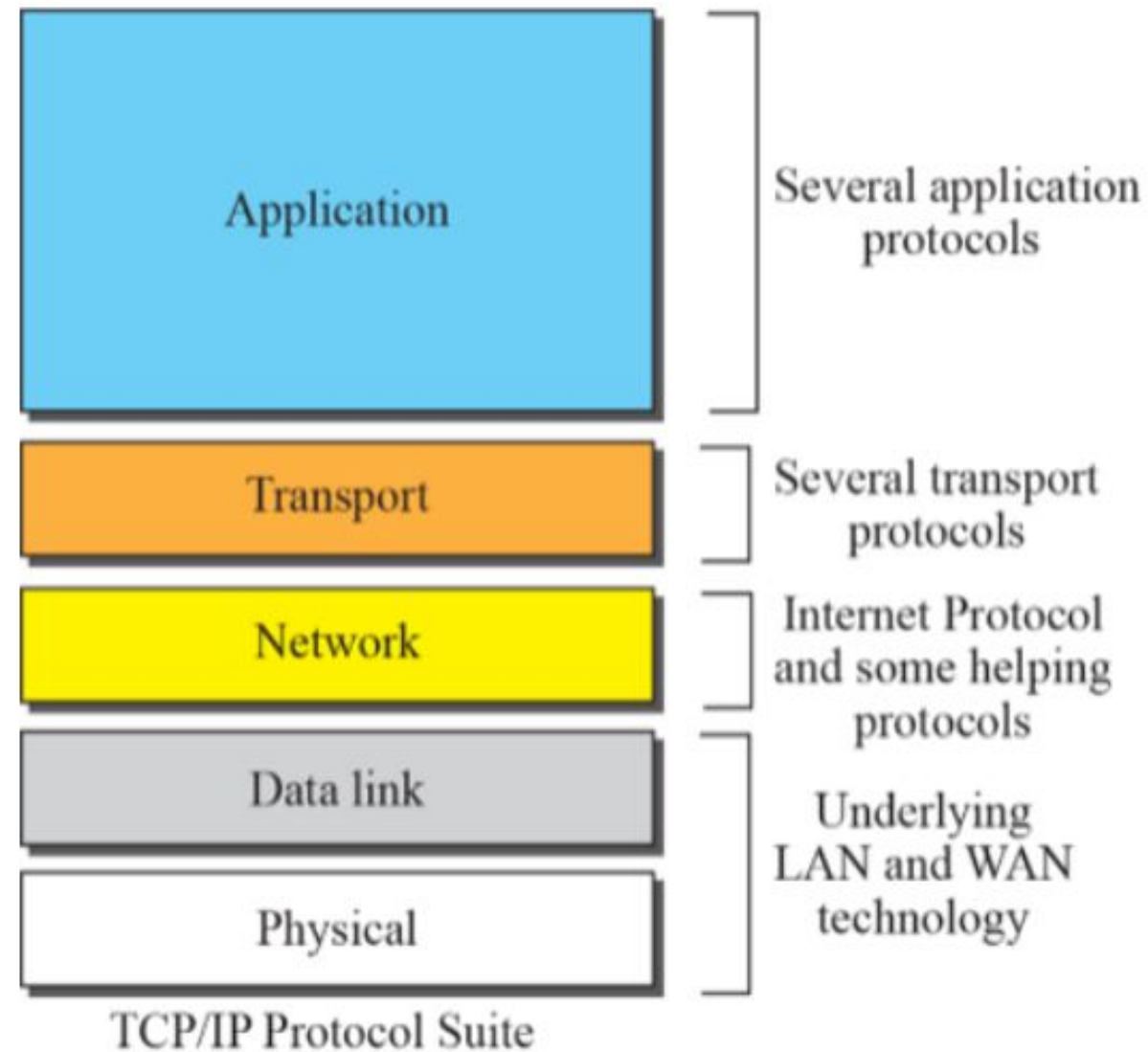
1.5 TCP/IP Model

- ❖ The TCP/IP protocol suite was developed prior to the OSI model.
- ❖ Therefore, the layers in the TCP/IP protocol suite do not match exactly with those in the OSI model.
- ❖ The original TCP/IP protocol suite was defined as four software layers built upon the hardware.
- ❖ Today, however, TCP/IP is thought of as a five-layer model with the layers named similarly to the ones in the OSI model.



1.5 TCP/IP Model

- ❖ The current Internet is based on a TCP/IP reference model.
- ❖ TCP and IP are two protocols of this model. TCP stands for Transmission Control Protocol and IP stands for Internet Protocol.
- ❖ The architecture or model was defined by the US department of defense and is used by ARPANET (Advanced Research Project Agency Network).



1.6 Physical Layer

- ❖ The Physical Layer decides and specifies exactly which types of cables, connectors, signal strengths, and transmission methods should be used so devices can communicate physically.
- ❖ The **lowest layer** of the OSI model (Layer 1).
- ❖ Directly interacts with the physical transmission medium.
- ❖ **Transmits raw bits (0s and 1s)** over the communication medium without interpreting their meaning.
- ❖ Converts data into **electrical signals, light pulses or radio waves** depending on the medium used.
- ❖ Defines **physical and electrical characteristics** of the interface between devices.
- ❖ The **Physical Layer** sets the **rules and standards** for how data is physically transmitted over a network.

1.6 Physical Layer

Example:

If two computers are connected by Ethernet, The following rules and standards are used

- **Cable type** → Cat6 twisted pair.
- **Connector type** → RJ-45.
- **Voltage levels** → 0.0 V for binary 0, 2.5 V for binary 1.
- **Wiring standard** → TIA/EIA-568B (defines which wire in the cable carries which signal).
- **Signal timing** → 1 bit every 100 nanoseconds for a 10 Mbps link.
- **Modulation** → Manchester encoding (how the voltage changes to represent bits).
- All these details are **defined by the Physical Layer** so that both ends can understand each other's signals.

1.6 Physical Layer

1. Bandwidth

- ❖ The maximum amount of data that can be transmitted over a network link or channel in a given amount of time.
- ❖ It shows the **capacity** of the connection.

Bandwidth in Hertz (Hz)

- ❖ In this sense, *bandwidth* means the **range of frequencies** that a channel or signal occupies. It is purely a physical measure of spectrum width, not the actual data rate.

Example:

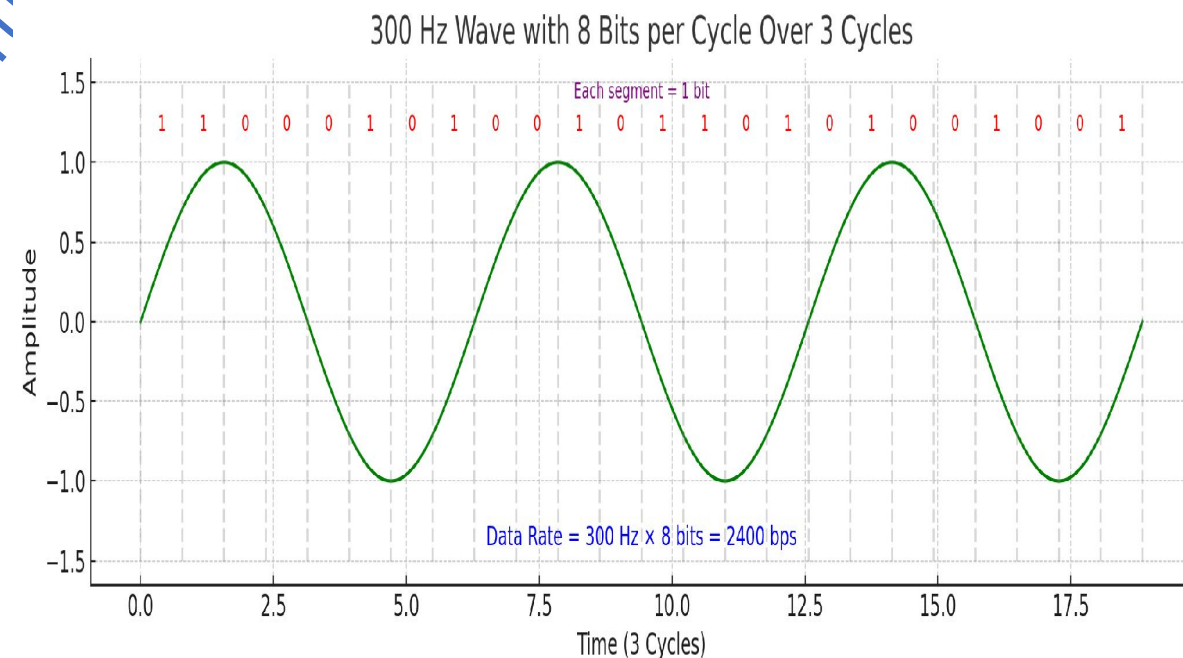
- ❖ A traditional analog telephone line can carry frequencies between **300 Hz and 3,400 Hz**.

- ❖ The range is: $3400 - 300 = 3100$ Hz (≈ 3.1 kHz)

- ❖ This measure matters because the **wider the frequency range**, the more “room” there is to carry information.

Real-life analogy:

- ❖ Think of bandwidth in Hz like the **width of a water pipe**—the wider the pipe, the more water (signal content) can potentially flow through.



1.6 Physical Layer

Bandwidth in Bits per Second (bps)

- ❖ Here, *bandwidth* means **data rate**—how many bits can be transmitted per second through a channel or link.
- ❖ Data Rate (bps)=Bandwidth (Hz)×Bits per Cycle (bps/Hz)

Example:

- ❖ A **Fast Ethernet** connection has a maximum bandwidth of **100 Mbps**.
This means it can transfer **100 million bits per second** in ideal conditions.

- ❖ **Real-life analogy:**

Think of bandwidth in bps like the **amount of water flowing per second**—not just the size of the pipe, but how fast and how much water is actually moving.

2. Latency or Delay:

- ❖ Latency is the total time it takes for a complete message to arrive at the destination, measured from the moment the first bit is sent from the source.
- It is made up of **four main components**:
 - i. **Propagation Time** – Time for a bit to physically travel from sender to receiver.
 - ii. **Transmission Time** – Time to push all bits of the message onto the link.
 - iii. **Queuing Time** – Time a message waits in buffer queues before transmission.
 - iv. **Processing Delay** – Time taken by devices (routers, switches, etc.) to examine and process the packet.

1.6 Physical Layer

1. Propagation Time

- Measures how long it takes for a signal to move from source to destination.

$\text{Propagation Time} = \text{Distance} / \text{Propagation Speed}$

2. Transmission Time

- Time to place all bits of a message onto the channel:

$\text{Transmission Time} = \text{Message Size (in bits)} / \text{Bandwidth (in bps)}$

3. Queuing Time

- Time the message spends **waiting in device queues** before being sent.

This depends on **network congestion**.

4. Processing Delay

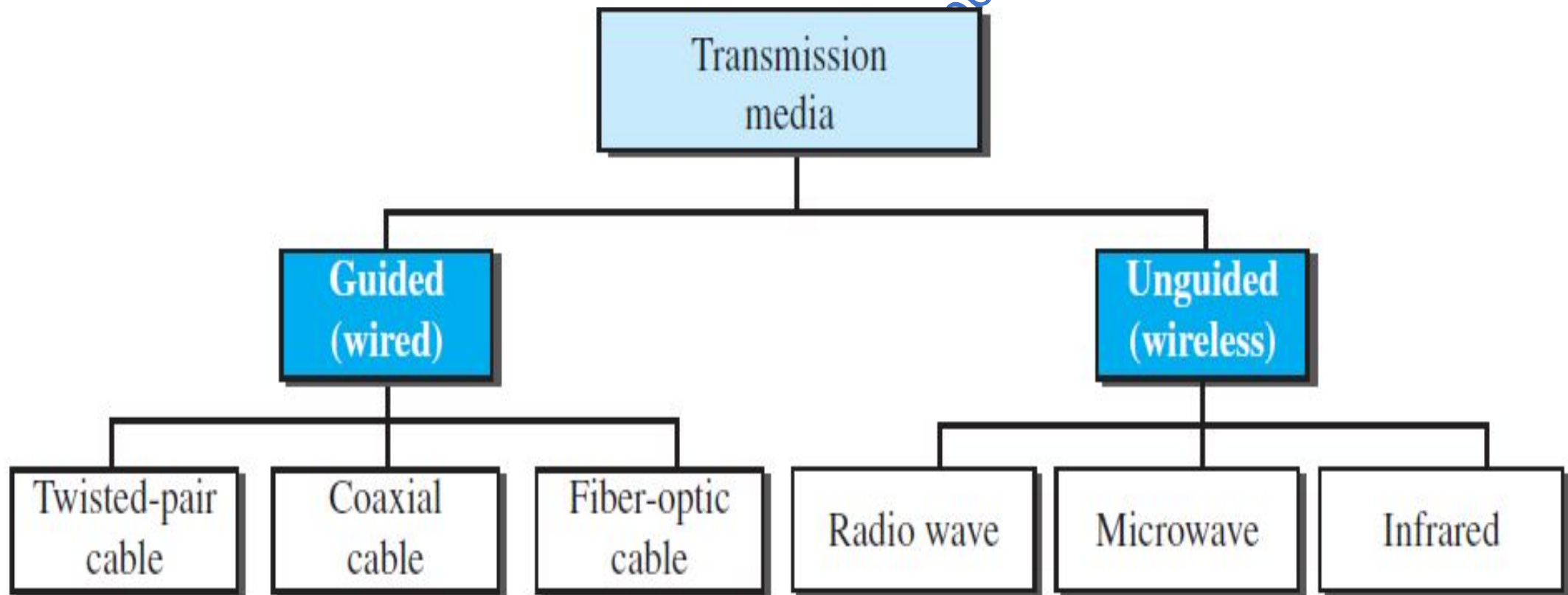
- Time taken for each intermediate device to **analyze headers, check errors, and decide forwarding**.

Total Latency = Propagation Time + Transmission Time + Queuing Time + Processing Delay

1.7 Guided Media

Transmission Medium

- A **transmission medium** is **any physical path or substance** that can carry information from a sender (source) to a receiver (destination).
- It can be **natural** (like air, water) or **man-made** (like cables, fiber optics).
- In telecommunications, transmission media can be divided into two broad categories: Guided and Unguided.



1.7 Guided Media

1. GUIDED MEDIA

- ❖ **Guided media** are physical transmission paths that provide a **channel** for signals to travel directly from one device to another.
- ❖ In these media, the signal is **directed and contained** within the physical boundaries of the cable.
- ❖ Types of Guided Media

- Twisted-Pair Cable**
- Coaxial Cable**
- Fiber-Optic Cable**

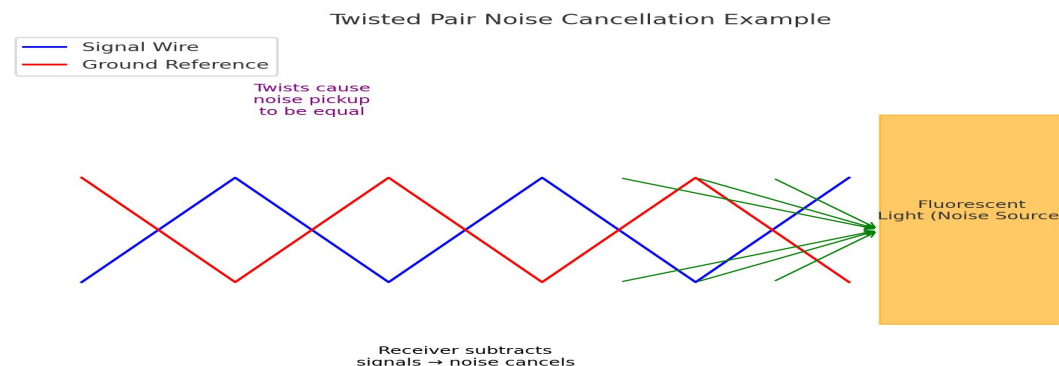
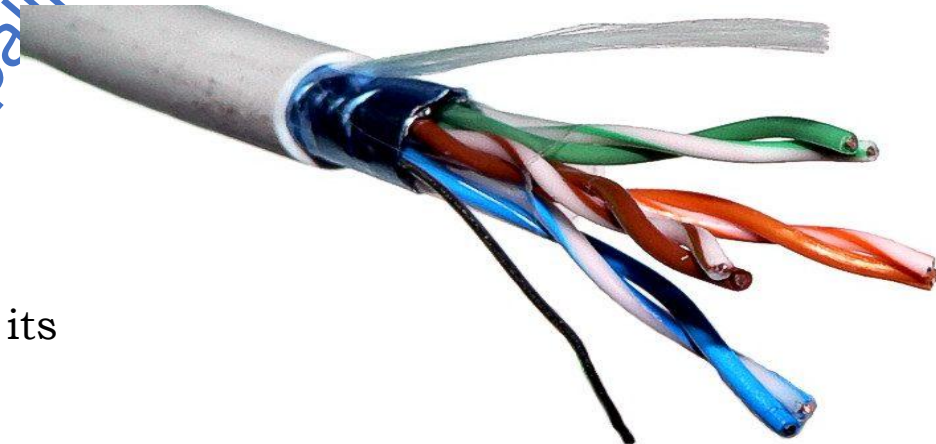
A. Twisted-Pair Cable

- ❖ A **twisted-pair cable** consists of **two copper conductors**, each with its own plastic insulation, twisted together.

- ❖ **One wire** carries the signal.

- ❖ **The other** serves as a ground reference.

- ❖ The receiver reads the **difference** between the two signals.



1.7 Guided Media

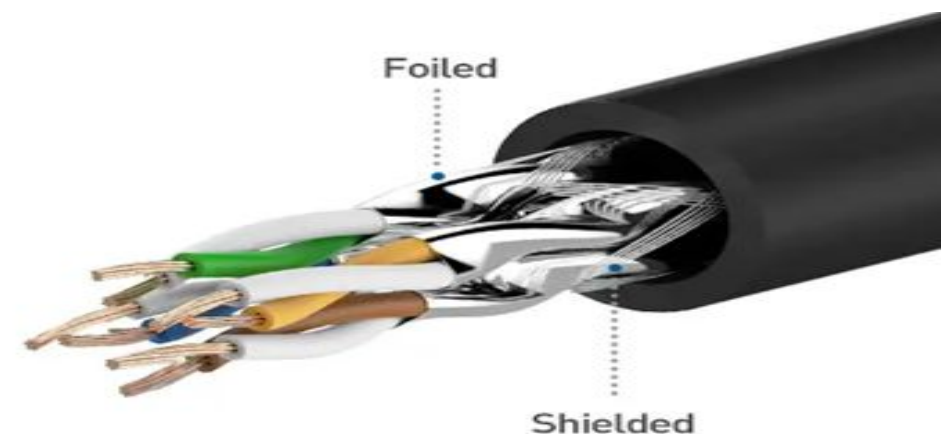
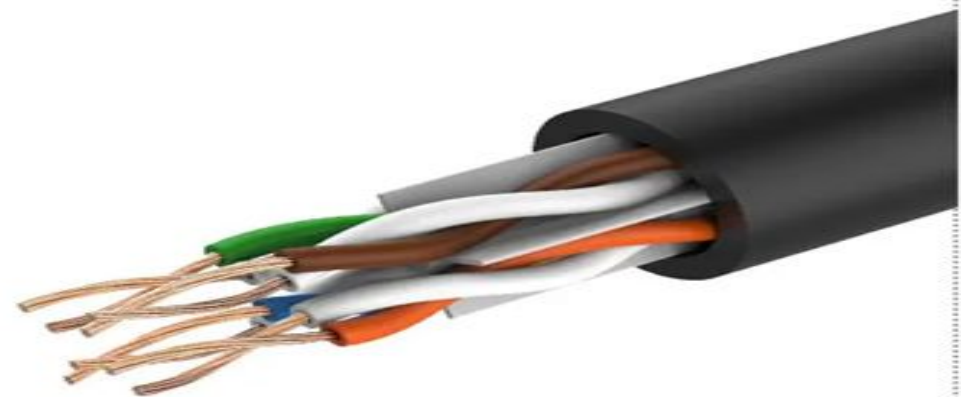
Types of Twisted-Pair Cable

- **Unshielded Twisted-Pair (UTP)**

- ☐ No extra shielding around the pairs.
- ☐ Most common in networking and telephone systems.
- ☐ Advantage: low cost, good flexibility, ease of installation
- ☐ Application: used in Ethernet networks, telephone systems, general data transmission

- **Shielded Twisted-Pair (STP)**

- ☐ Each pair is covered with a metal foil or braided mesh to reduce noise and crosstalk.
- ☐ Common in IBM systems; rarely used outside.
- ☐ Advantage: better anti-interference performance, higher signal integrity
- ☐ Application: used in environments sensitive to electromagnetic interference, such as industrial control systems, medical equipment, and high-speed data transmission



1.7 Guided Media

Connectors are the physical plugs and sockets used to attach your twisted-pair cable to devices like computers, routers, sensors, or switches.



RJ - 45 Male



RJ - 45 Female

Applications:

1. Telephone Networks
2. Local Area Networks (LANs)
3. DSL Internet Connections
4. Security and CCTV Systems
5. Intercom Systems

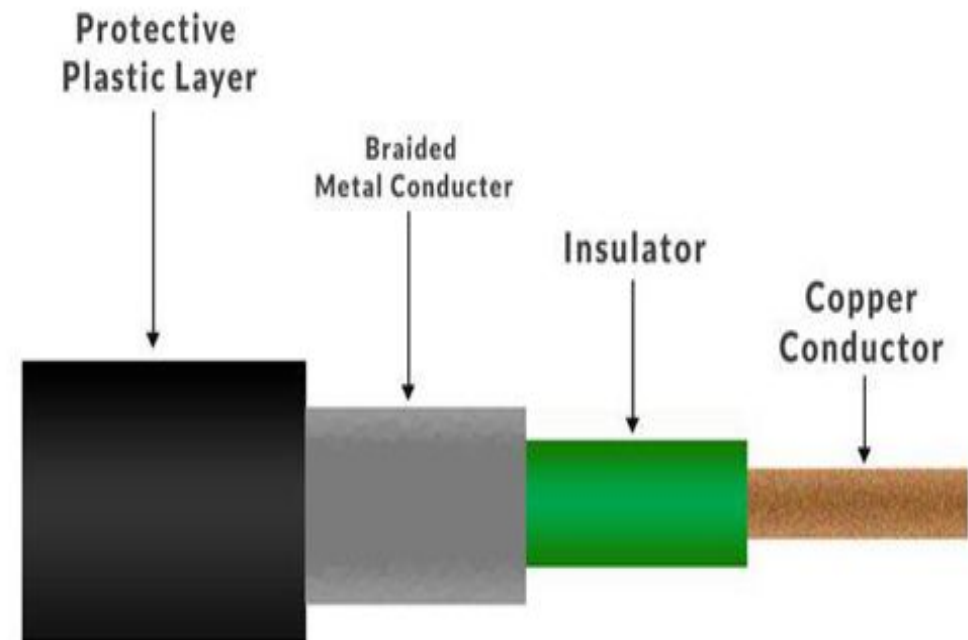
1.7 Guided Media

B. Coaxial Cable

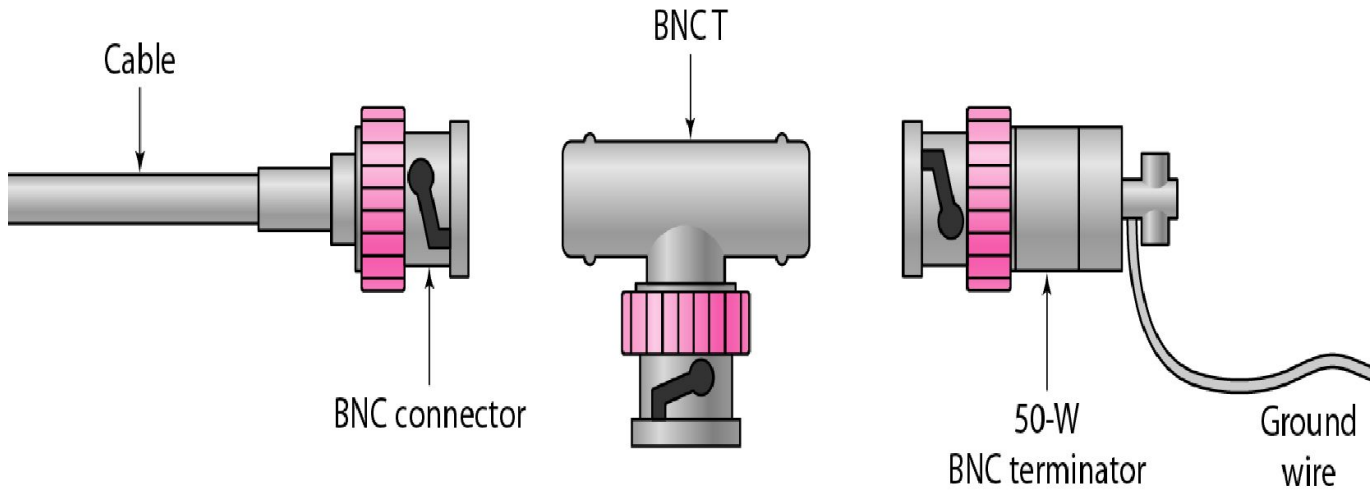
- ❖ Coaxial Cable is a type of guided media made of Plastics, and copper wires which transmit the signal in electrical form rather than light form.
- ❖ Coaxial cable is also known as coax. The core copper conductor is used for the transmission of signals and the insulator is used to provide insulation to the copper conductor the insulator is surrounded by a braided metal conductor which helps to prevent the interference of electrical signals and prevent cross talk.
- ❖ This entire setup is again covered with a protective plastic layer to provide extra safety to the cable.

Structure of Coaxial Cable

- ❑ **Copper conductor:** A central conductor, which consists of copper. The conductor is the point at which data is transmitted.
- ❑ **Insulator:** Dielectric plastic insulation around the copper conductor. it is used to maintain the spacing between the center conductor and shield.
- ❑ **Braided mesh:** A braided mesh of copper helps to shield from electromagnetic interference, The braid provides a barrier against EMI moving into and out of the coaxial cable.
- ❑ **Protective plastic layer:** An external polymer layer, which has a plastic coating. It is used to protect internal layers from damage.



1.7 Guided Media



Advantages of Coaxial Cable

- ❖ Coaxial cables support high bandwidth.
- ❖ It is easy to install coaxial cables.
- ❖ Less affected by noise or cross-talk or electromagnetic inference.
- ❖ Coaxial cables support multiple channels

Applications of Coaxial Cable

- ❖ Television
- ❖ Internet
- ❖ CCTV
- ❖ Video
- ❖ HDTV

Disadvantages of Coaxial Cable

- ❖ Coaxial cables are expensive.
- ❖ The coaxial cable must be grounded in order to prevent any crosstalk.
- ❖ As a Coaxial cable has multiple layers it is very bulky.
- ❖ There is a chance of breaking the coaxial cable and attaching a "t-joint" by hackers, this compromises the security of the data.

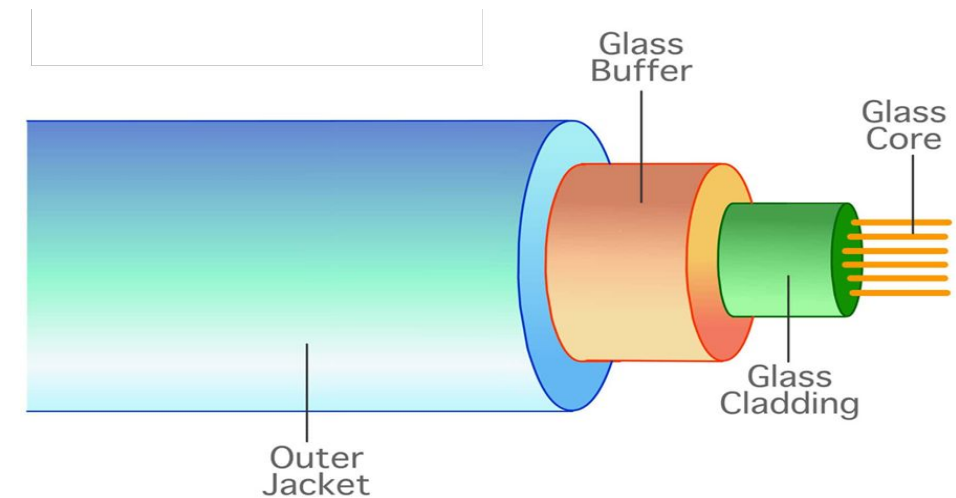
1.7 Guided Media

C. Fiber Optic Cable

- ❖ A fibre optic cable is made of glass or plastic and transmits signals in the structure of light signals.
- ❖ In fiber optics, semiconductor lasers are used to generate light signals that carry data.
- ❖ These light pulses travel through hair-thin glass or plastic fibers—called optical fibers—at extremely high speeds (up to the speed of light in the medium, which is about 186,000 miles per second in a vacuum, slightly slower in glass). The technology allows data to be transmitted over very long distances with minimal loss of signal strength, making it highly efficient for communication systems.

Structure of Coaxial Cable

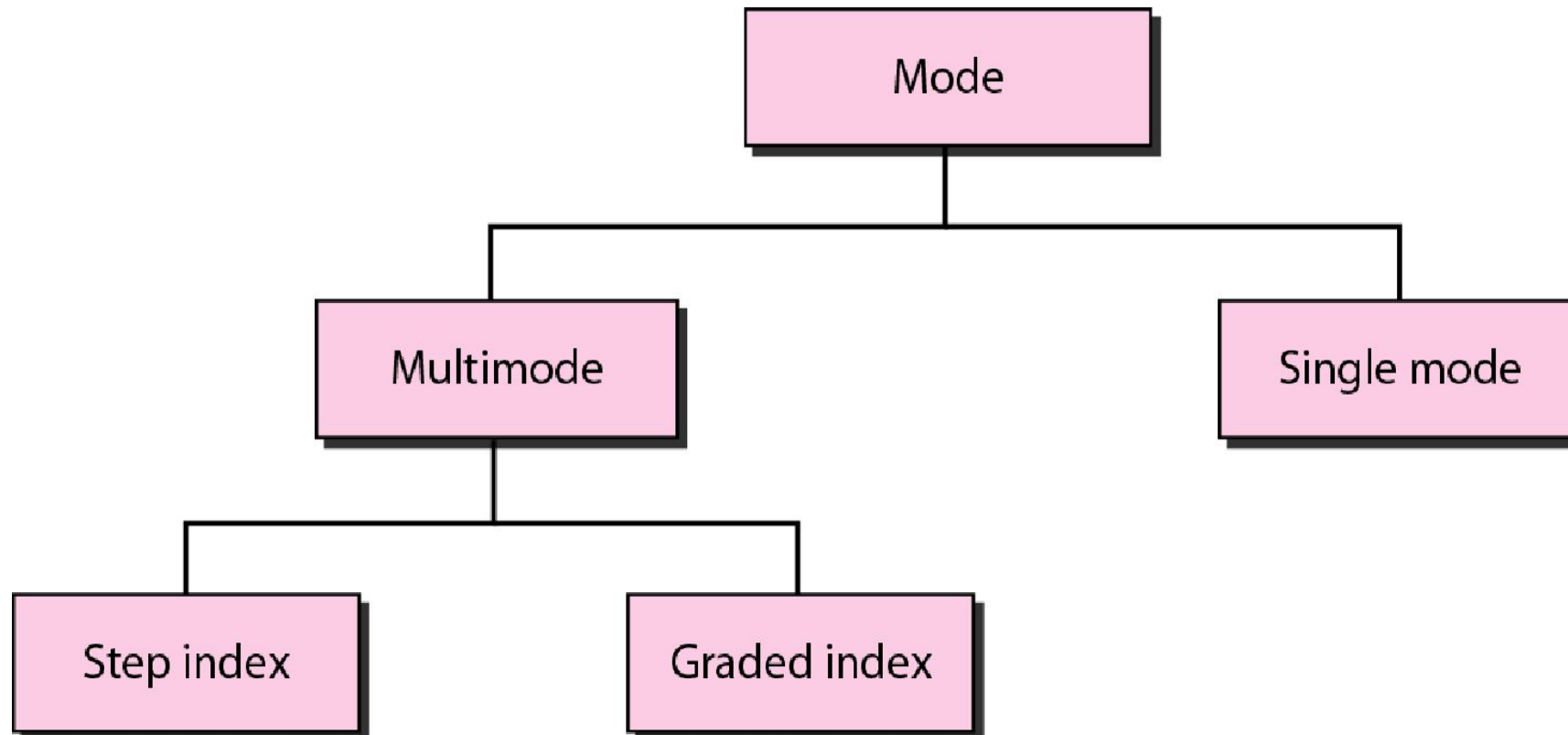
- ❖ **Outer Jacket** – The protective outer covering that shields the cable from environmental damage, moisture, and physical wear.
- ❖ **Glass Buffer** – A protective layer that cushions the inner fiber, preventing breakage and providing mechanical strength.
- ❖ **Glass Cladding** – A layer of glass with a lower refractive index than the core, causing light to reflect back into the core (total internal reflection) and keep signals traveling inside.
- ❖ **Glass Core** – The central part of the fiber where light signals actually travel, made of high-purity glass or plastic for efficient signal transmission.



1.7 Guided Media

C. Fiber Optic Cable

Propagation modes



1.7 Guided Media

C. Fiber Optic Cable

Multimode –Step index

- **Multimode is so named because multiple beams from a light source move through the core in different paths. How these beams move within the cable depends on the structure of the core,**

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1.7 Guided Media

Applications of Fiber Optic Cable

- ☐ High-speed broadband connections
- ☐ Data Centres & Networking
- ☐ Cable Television (CATV)
- ☐ Military & Aerospace
- ☐ Medical
- ☐ Sensing & Monitoring

Advantages of Fiber Optic Cable

- ☐ Higher bandwidth
- ☐ Less signal attenuation
- ☐ Immunity to electromagnetic interference
- ☐ Resistance to corrosive materials
- ☐ Light weight
- ☐ Greater immunity to tapping

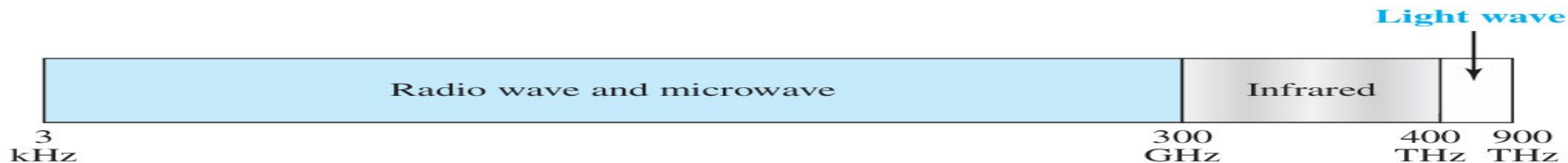
Disadvantages of Fiber Optic Cable

- ☐ Installation and maintenance
- ☐ Unidirectional light propagation
- ☐ Cost

1.7 Unguided Media

UNGUIDED MEDIA: WIRELESS

- ❑ Unguided medium transport electromagnetic waves without using a physical conductor.
- ❑ This type of communication is often referred to as wireless communication.
- ❑ Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.
- ❑ Unguided mediums are ranging from 3 kHz to 900 THz, used for wireless communication.
- ❑ Unguided signals can travel from the source to the destination in several ways: ground propagation, sky propagation, and line-of-sight propagation



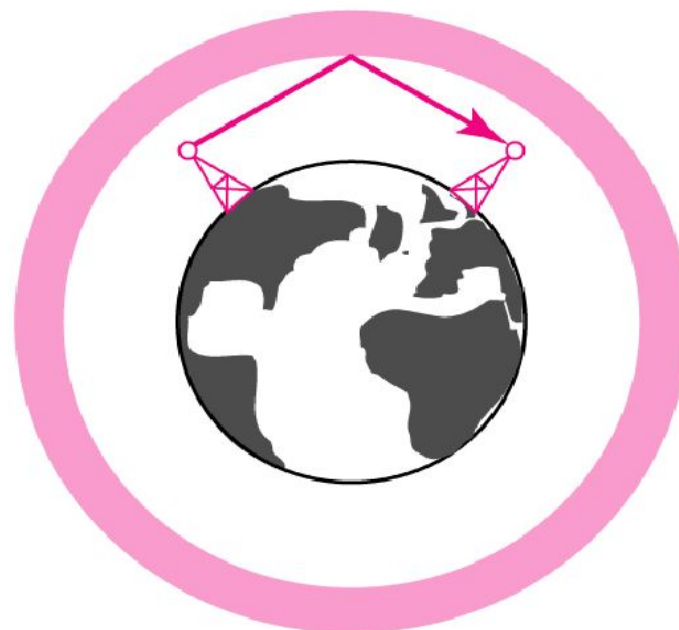
1.7 Unguided Media

Ionosphere



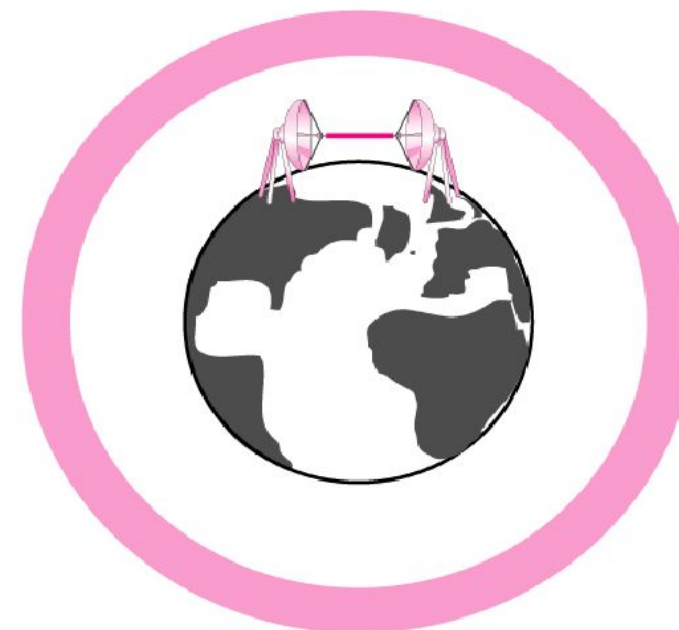
Ground propagation
(below 2 MHz)

Ionosphere



Sky propagation
(2–30 MHz)

Ionosphere



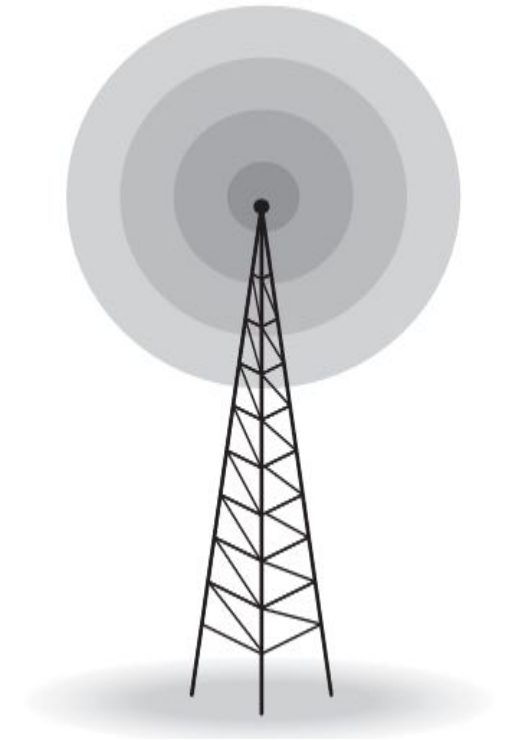
Line-of-sight propagation
(above 30 MHz)

1 Hz → 1 wave cycle per second
10 Hz → 10 wave cycles per second
1 kHz (kilohertz) → 1,000 cycles per second
1 MHz (megahertz) → 1,000,000 cycles per second
1 GHz (gigahertz) → 1,000,000,000 cycles per second

1.7 Unguided Media

1. Radio Waves

- ❖ Radio waves are electromagnetic waves that typically occupy the **3 kHz to 1 GHz** frequency range, with microwaves above 1 GHz up to 300 GHz.
- ❖ Their most distinctive property is **omnidirectional propagation**, meaning they radiate in all directions from a transmitting antenna, allowing communication without precise alignment between sender and receiver.
- ❖ They can **travel long distances**, especially via sky-wave propagation, making them useful for broadcasting applications like AM radio.
- ❖ Their ability to **penetrate walls** enables reception indoors, but this also means signals cannot be easily confined to a specific area.
- ❖ The omnidirectional property has a disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band.
- ❖ Radio waves, particularly those waves that propagate in the sky mode, can travel long distances. This makes radio waves a good candidate for long-distance broadcasting such as AM radio. Radio waves, particularly those of low and medium frequencies, can penetrate walls.

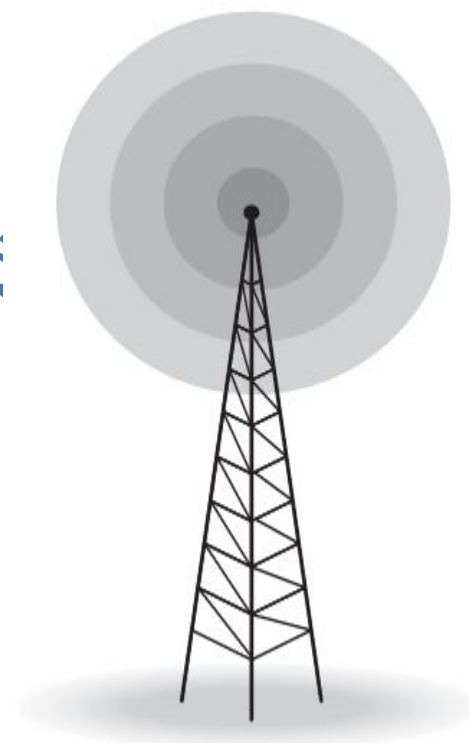


1.7 Unguided Media

Applications of Radio Waves

- ❖ AM and FM radio
- ❖ Television
- ❖ Maritime radio
- ❖ Cordless phones
- ❖ Paging

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1.7 Unguided Media

2. Microwaves

- ❖ Electromagnetic waves having frequencies between 1 and 300 GHz are called micro waves. Microwaves are unidirectional.
- ❖ When an antenna transmits microwaves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.
- ❖ The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas.
- ❖ **The following describes some characteristics of microwave propagation:**
 - ❑ Microwave propagation is line-of-sight. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall.
 - ❑ Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside buildings.
 - ❑ The microwave band is relatively wide, almost 299 GHz. Therefore, a high data rate is possible.
 - ❑ Use of certain portions of the band requires permission from authorities.

1.7 Unguided Media

2. Microwaves

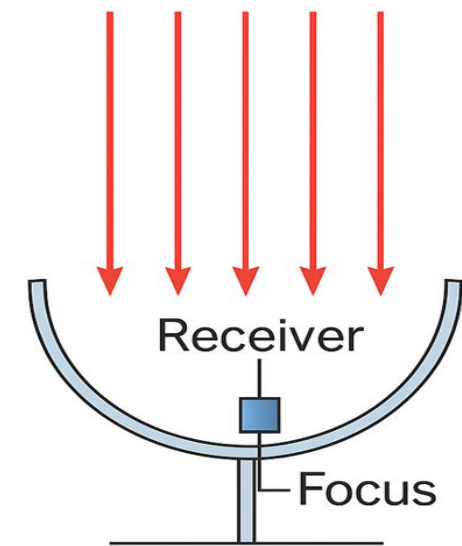
Unidirectional Antenna Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used for microwave communications: **the parabolic dish and the horn.**

1. Parabolic dish

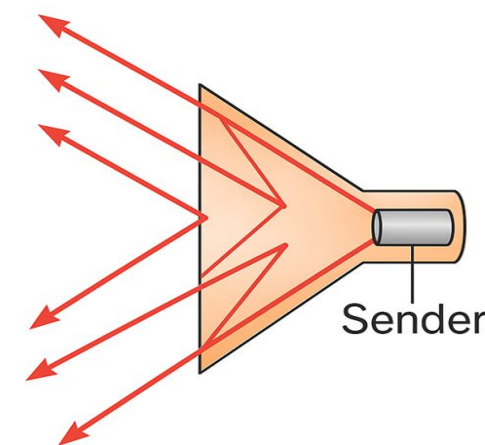
- ❑ A **parabolic dish antenna** uses the shape of a parabola to focus electromagnetic waves. Any incoming wave parallel to its line of symmetry reflects toward a single point called the **focus**, where the receiver is placed.
- ❑ This design works like a funnel, capturing a wide range of incoming signals and directing them to one point, thereby increasing signal strength compared to a single-point receiver. For transmission, a feed horn sends microwaves toward the dish, which reflects them outward as a focused beam.

2. Horn

- ❑ A horn antenna resembles a large scoop. In transmission, signals travel through a narrow waveguide (the “stem”) and spread outward as parallel beams when they exit the flared “horn.”
- ❑ For reception, the horn collects incoming waves and directs them down the waveguide to the receiver. Like a parabolic dish, it focuses signals but uses its flared shape instead of a curved reflector.



Parabolic Dish

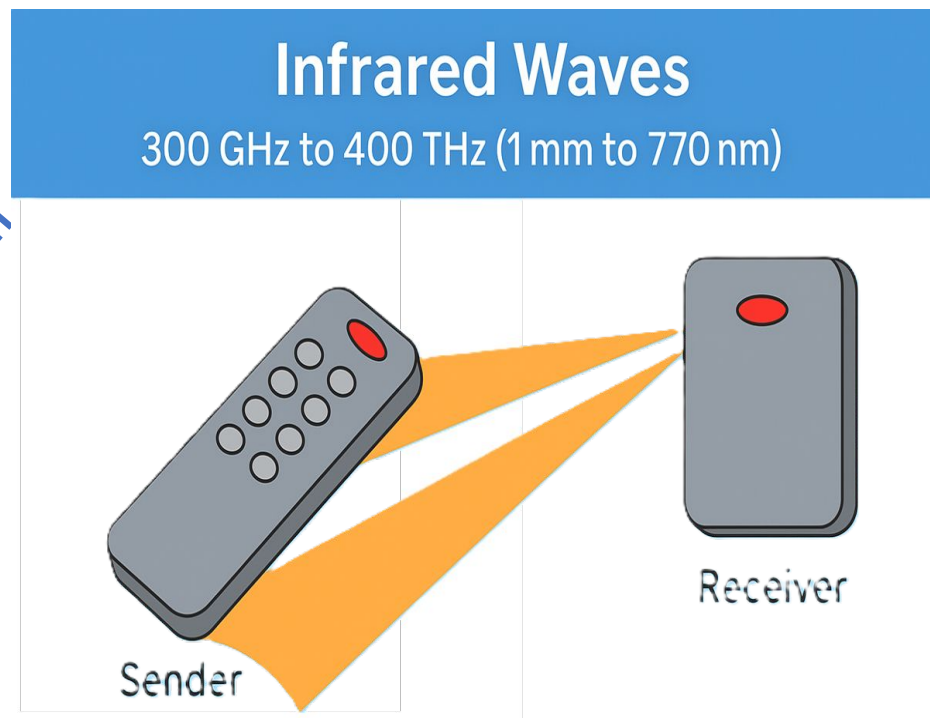


Horn Antenna

1.7 Unguided Media

3. Infrared

- ❖ Infrared waves occupy the frequency range from **300 GHz to 400 THz**, with wavelengths between **1 mm and 770 nm**. They are primarily used for **short-range communication** due to their high frequency.
- ❖ A key property of infrared waves is that they **cannot penetrate walls**. This is advantageous because it prevents interference between systems in different rooms — for example, using an infrared remote control in one room will not affect a similar device in another.
- ❖ However, this same limitation makes infrared waves **unsuitable for long-range communication**, as the signals are easily blocked by obstacles and cannot travel over large distances.



1.7 Unguided Media

Applications of Infrared Band (IrDA):

- ❑ **Wireless computer peripherals** – Connects keyboards, mice, and printers to PCs without cables.
- ❑ **Short-range device communication** – Enables point-to-point data transfer between compatible devices.
- ❑ **Office automation** – Facilitates wireless printing and file sharing.
- ❑ **Consumer electronics** – Used in remote controls and other control devices.
- ❑ **Secure line-of-sight links** – Prevents interference with devices in other rooms.

